

Structural Hawkes Ambiguity in Near-Critical Market Making

Abstract

This paper studies a narrow source of model risk in market making: uncertainty about the self-excitation structure of Hawkes order flow. The structure is encoded by a branching matrix Γ , while the distance to instability is measured by the Perron root $\rho(\Gamma)$. In a reduced robust inventory model, ambiguity in Γ is amplified through the Hawkes resolvent $(I - \Gamma)^{-1}$, so near-critical calibration error can become an economic risk premium. The critical exponent depends on normalization: relative critical-slack ambiguity gives a square-law reduced premium, while absolute branching-matrix ambiguity contributes one additional derivative factor when $\epsilon = o(1 - \rho)$.

The paper gives a reduced-form spectral theorem and conditional robust-control bridges under stated compact-PDMP, Lyapunov, weighted-comparison, and smooth-interior assumptions. It then evaluates robust and nominal quoting policies under common random numbers, Ogata event times, public LOBSTER sample replays, Binance aggregate trades, and crypto order-book samples. The empirical claim is deliberately limited: structural Hawkes ambiguity is visible in these near-critical stress tests, but public Hawkes calibrations remain diagnostic inputs rather than complete execution models.

Author. Mayank Sharma, Indian Institute of Technology Jodhpur, Department of Engineering Science. Contact: may1729int@gmail.com.

Status. Preprint / working paper.

Keywords. Hawkes processes; robust market making; structural ambiguity; near-critical order flow; viscosity solutions; limit order books; queue-position risk.

AI disclosure. Generative AI tools assisted with language editing, LaTeX cleanup, code scaffolding, experiment automation, validation-script review, and reproducibility-package organization. The author reviewed the outputs and remains responsible for all mathematical claims, proofs, citations, code, data processing, numerical experiments, and final text.

Reproducibility package. Code, tests, TeX sources, derived tables, and derived figures: github.com/mayank172900/mesa-sifin-reproducibility.

1 Introduction

When liquidity disappears during stress, the usual explanations are volatility, inventory pressure, or lower fill probabilities. This paper focuses on a smaller but sharper mechanism. If order flow is self-exciting and already close to criticality, then uncertainty about the excitation structure can become expensive by itself. The same estimation error in Γ may be harmless far from criticality and serious when $\rho(\Gamma)$ is close to one.

The contribution is narrow by design. Hawkes market making, robust market making, near-critical Hawkes limits, and learned limit-order-book simulators are all active areas. This paper studies their intersection: structural Hawkes ambiguity in market making, the spectral amplification law it creates, and reproducible diagnostics showing why near-critical Hawkes calibrations should be stress-tested before being used inside a trading simulator.

The paper makes four contributions. It identifies the resolvent channel through which branching-matrix uncertainty amplifies risk-premium proxies. It separates relative critical-slack ambiguity

from absolute branching-matrix ambiguity, which changes the critical exponent. It develops the reduced control problem through a sequence of conditional finite-state, compact-PDMP, Lyapunov, weighted-comparison, and interior-optimizer results. Finally, it reports a reproducible experimental path from synthetic scaling tests to robust Bellman policies, event-time execution stress tests, public LOBSTER replays, Binance aggregate-trade diagnostics, and crypto depth sanity checks.

Reader’s map

The main idea can be understood before reading the mathematical appendix that follows the paper in the combined SSRN PDF. If order flow is described by a Hawkes branching matrix Γ , then the long-run effect of arrivals is filtered through the resolvent $(I - \Gamma)^{-1}$. As $\rho(\Gamma)$ moves toward one, this resolvent grows in the leading Perron direction. For a market maker uncertain about the excitation matrix, this makes near-critical calibration error more than a small nuisance parameter. It can be magnified into a meaningful economic risk.

The rest of the paper proceeds in three layers. The first is spectral and gives the normalization-dependent amplification laws. The second is a robust-control bridge showing how the spectral effect enters reduced Bellman/HJBI equations for market making. The third is empirical and computational. The experiments do not form a production execution engine. They ask whether the same risk layer remains visible under finite-state policies, event-time simulation, public LOBSTER replay, crypto clustering diagnostics, calibration noise, and ablation tests.

Main claim. MESA—market-making excitation-structure ambiguity—does not claim that Hawkes models fully describe limit-order-book execution. It says that, conditional on using a Hawkes structural layer, ambiguity in Γ is amplified near criticality, and the amplification exponent depends on how that ambiguity is normalized. The theory establishes this reduced-form spectral claim under stated assumptions; the experiments test whether the mechanism is visible in the stress designs reported below.

2 Related Work and Positioning

Nearby literatures approach this problem from several directions. Point-process and Hawkes market-making models study weakly consistent LOB control [2], Hawkes impulse control [10], and adversarial or reinforcement-learning variants [6]. A separate LOB-simulation and forecasting literature uses neural Hawkes, multivariate Hawkes, state-dependent Hawkes, and non-Hawkes realism-focused simulators to improve realism and calibration [7, 3, 4, 11, 12, 13]. Recent work on robust quote refusal and latency-aware policies addresses different forms of execution risk [5, 9]. Empirical Hawkes analyses also motivate concern about near-critical financial event flow [16], while nearly unstable Hawkes limits clarify what happens close to criticality [14, 15]. MESA is narrower than these programs. Throughout the paper, MESA means market-making excitation-structure ambiguity: a spectral ambiguity layer for Hawkes branching matrices. It does not propose a new full LOB simulator or a competing RL agent; it isolates how uncertainty in the branching matrix is amplified when the Hawkes system is near instability.

The reproducibility package records the source URLs, visible code links, and implementation notes used to construct the comparison. The intended position is therefore modest: MESA is a spectral ambiguity-risk layer that can be attached to richer simulators, not a substitute for them.

Work	Hawkes MM	Robust MM	Γ uncertainty	Near-critical scaling	MESA position
Law–Viens 2019/2020	no	no	no	no	Marked point-process/HJB-QVI LOB MM benchmark; MESA adds Hawkes branching-matrix ambiguity.
Wang–Ventre–Polukarov 2023/2025 quote/no-quote	no	yes	no	no	Robust quote-refusal/single-sided quoting benchmark; Hawkes branching uncertainty is outside its scope.
Wang–Ventre–Polukarov 2024/2025 Hawkes ARL	yes	yes	no	no	A close Hawkes/ARL policy benchmark; no branching-matrix ambiguity amplification law.
Jain et al. 2025 Hawkes impulse control	yes	no	no	no	A close HJB-QVI benchmark; robustness to Γ is not the main object.
Lalor–Swishchuk 2025 neural Hawkes LOB	yes	no	no	no	Neural Hawkes LOB simulator/DRL market-making baseline; ambiguity amplification is complementary.
Kumar 2021/2024 Deep Hawkes	yes	no	no	no	Deep Hawkes high-frequency MM/simulation benchmark; MESA adds structural Γ ambiguity diagnostics.
Guo–Lin–Huang 2023 Attn-LOB DRL [8]	no	no	no	no	LOB-feature DRL baseline; focus is alpha learning rather than Hawkes structure.
Jiang et al. 2025 latency-aware RL	no	no	no	no	Latency/inventory-aware neural MM benchmark; robust Hawkes ambiguity is not modeled.
Raffaelli et al. 2026 multivariate Hawkes BTC LOB	no	no	no	no	Recent multivariate Hawkes BTC LOB calibration/forecasting study; robust control is outside its scope.
El Karmi 2025 deterministic Hawkes LOB simulator	no	no	no	no	Simulator-realism baseline with Hawkes-driven order flow; robustness is complementary.
Noble–Rosenbaum–Souilmi 2026 realism gap	no	no	no	no	Recent simulator-realism and execution-sensitivity study; MESA is a complementary risk layer.
Kimura 2026 state-dependent Hawkes	no	no	no	no	State-dependent Hawkes LOB with local supercriticality evidence; MESA targets robust Γ ambiguity.
Szymanski–Xu 2025 nearly unstable Hawkes	no	no	no	yes	Near-critical Hawkes scaling limits; MESA adds market-making control and Γ -ambiguity premiums.
El Karmi 2026 bivariate nearly unstable Hawkes	no	no	no	yes	Recent bivariate near-critical Hawkes limit theory; MESA targets robust market-making premiums.
MESA reduced-form package	yes	yes	yes	yes	Reduced-form theorem, policy stress tests, and public-data calibration diagnostics.

Table 1: Comparison with nearby work. The point is not to claim simulator primacy, but to isolate the structural Hawkes ambiguity layer.

3 Model

Let $N_t^{k,s}$ denote marked order arrivals by type k and side s . The exponential Hawkes intensity is

$$\lambda_t^{k,s} = \mu^{k,s} + \sum_{j,\sigma} \int_0^{t-} \alpha_{j,\sigma}^{k,s} \beta e^{-\beta(t-u)} dN_u^{j,\sigma}.$$

Thus Γ records the integrated excitation from each event type into each other event type. Stability is governed by its Perron root: when $\rho(\Gamma) < 1$, the stationary mean intensity is

$$\bar{\lambda} = (I - \Gamma)^{-1} \mu.$$

This resolvent is the channel through which small errors in Γ become large near criticality.

A market maker chooses bid and ask offsets $u_t = (\delta_t^b, \delta_t^a) \in \mathcal{U}$ with inventory $q_t \in \{-Q, \dots, Q\}$. Nature chooses a structural model from an ambiguity set

$$\mathcal{A}_\epsilon = \{\Gamma \geq 0 : \|\Gamma - \hat{\Gamma}\|_F \leq \epsilon, \rho(\Gamma) \leq \rho_{\max} < 1\}.$$

The asymptotic statements are interpreted along Perron-regular local sequences with $\rho_{\max}$ allowed to approach one, while numerical experiments use finite stable stress scenarios below one. For absolute ambiguity, the derivative law is local: if ϵ is not smaller than the remaining critical slack, the perturbation can saturate at the stability cap and the simple $(1 - \rho)^{-3}$ expansion no longer applies. For numerical work, this ambiguity set is discretized into the nominal matrix and a small collection of Perron-aligned boundary perturbations.

4 Theory and Main Results

Assumption 1 (Perron regularity). Γ is nonnegative, irreducible, and primitive, with Perron root $\rho < 1$. Let $\text{gap}(\Gamma) = \rho - \max_{j>1} |\lambda_j(\Gamma)|$. Assume $\text{gap}(\Gamma) \geq g_0 > 0$, the Perron root is simple, the Perron eigenprojection $r\ell^\top$ is uniformly bounded under the normalization $\ell^\top r = 1$, and the reduced resolvent on the non-Perron spectral subspace is uniformly bounded. This rules out non-Perron conditioning from carrying the same singular order as the Perron pole.

Proposition 1 (Resolvent amplification). Under Perron regularity,

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho(\Gamma)} + O(1/\text{gap}).$$

Consequently, price-risk loadings whose Perron projection is bounded away from zero on the Perron-regular class are exposed to the same leading near-critical blow-up.

Proposition 2 (Perturbation direction). For a small perturbation E ,

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2 / \text{gap}).$$

The first-order Frobenius direction that maximally increases the Perron root is $E^* = \ell r^\top / \|\ell r^\top\|_F$, distinct from the resolvent projection $r\ell^\top$.

Theorem 1 (Reduced-form normalization-dependent ambiguity premium). Suppose the robust half-spread correction is locally linear in an effective Hawkes variance proxy proportional to $(1 - \rho)^{-2}$. The square-law exponent uses this normalized-throughput/resolvent-squared risk proxy; alternative

raw count-variance conventions may change the baseline exponent, but not the distinction between relative-slack and absolute- Γ ambiguity. Then relative critical-slack ambiguity, $\Delta\rho = O(\epsilon_{\text{rel}}(1 - \rho))$, gives

$$\psi = O\left(\frac{\epsilon_{\text{rel}}}{(1 - \rho)^2}\right).$$

Absolute branching-matrix ambiguity, $\Delta\rho = O(\epsilon)$ with $\epsilon = o(1 - \rho)$, gives the more singular derivative law

$$\psi = O\left(\frac{\epsilon}{(1 - \rho)^3}\right).$$

Matching lower bounds require Perron-visible loadings, a feasible Perron-aligned perturbation, an active worst-case selector, and an interior quote regime before spread caps or no-quote constraints bind.

Remark 1. The theorem is stated at the reduced-form level because the amplification mechanism is spectral, while the market-making problem can be embedded in several control formulations. The mathematical appendix gives scalar and finite-dimensional Perron-visible versions of the bridge, a finite-state Bellman verification theorem, and a compact continuous-intensity HJBI verification theorem for a truncated multitype Hawkes control game. It also records the Lyapunov, weighted-comparison, and local differentiability arguments used to connect the reduced premium calculation to smooth interior quote optimizers. The main limitations are active-set changes from no-quote or spread caps, the compact-PDMP comparison hypotheses, and extensions to state-dependent, power-law, or learned Hawkes kernels.

5 Computational Design

The empirical design combines complementary simulations, control solvers, and replay diagnostics. Its roles are to test the predicted critical scaling, evaluate robust quote policies under controlled Hawkes variation, and compare the resulting behavior with public limit-order-book data. The components are:

1. Scaling diagnostics for scalar and multitype resolvent proxies.
2. Common-random-number policy evaluation under full simulated Hawkes paths.
3. A finite-scenario robust inventory Bellman solver for a scalar reduced state with side-level quote/no-quote actions.
4. A reduced event-queue backtest on Ogata event times with queue-ahead mechanics and no-quote side-time accounting.
5. Displayed-depth, residual-volume priority-aware level-10, and deepest-public level-50 LOBSTER replays on public message/order-book streams.
6. Observable level-10 order-book reconstruction from messages, checked against official LOBSTER snapshots.
7. Ogata-thinning validation and time-step bias diagnostics.
8. LOBSTER Hawkes diagnostics over event type, time scale, marks, and side labels.
9. Binance aggregate-trade event diagnostics for public crypto clustering.

The Bellman recursion uses side-specific quote/no-quote controls: on each side the market maker

either posts at a chosen offset or withdraws liquidity.

$$h_n(q) = \max_{a^b, \delta^b, a^a, \delta^a} \min_{\Gamma \in S} \left\{ -\Delta t \left(\phi q^2 + \frac{\gamma}{2} \sigma^2(\Gamma) q^2 \right) \right. \\ \left. + p_b(\delta^b + h_{n+1}(q+1)) + p_a(\delta^a + h_{n+1}(q-1)) \right. \\ \left. + (1 - p_b - p_a)h_{n+1}(q) \right\}.$$

The indicators $a^b, a^a \in \{0, 1\}$ encode side activity; when a side is inactive, its fill probability is set to zero.

Theorem 2 (Finite-state Bellman verification). *Consider a finite state space encoding inventory and Hawkes regimes, a nonempty finite quote/no-quote action set at each state, a conservative finite-state generator $L^{u,\Gamma}$, bounded rewards, bounded terminal payoff, and a finite rectangular ambiguity set S of stable branching matrices. The finite-regime robust control problem has the max–min value. In continuous time its value is the unique bounded solution of the finite-state HJBI ODE*

$$-\partial_t V(t, x) = \max_{u \in U(x)} \min_{\Gamma \in S} \left\{ r(x, u, \Gamma) + \sum_y L^{u,\Gamma}(x, y) V(t, y) \right\}, \quad V(T, x) = g(x),$$

On a time grid, the analogous backward Bellman recursion is exact for the chosen stochastic one-step kernel, and for an explicit Euler kernel requires Δt small enough that $I + \Delta t L^{u,\Gamma}$ is stochastic. Deterministic Markov selectors attain the finite max–min. This identifies the finite-scenario quote/no-quote dynamic program with a discretized robust finite-state game. The continuous-intensity Hawkes HJBI is not covered by this finite-state argument; the compactness, Lyapunov, and weighted-comparison results below supply the bridge to that setting.

Theorem 3 (Conditional compact continuous-intensity HJBI verification). *Consider the truncated multitype Hawkes control state $X_t = (q_t, \lambda_t)$, with finite inventory $q_t \in \{-Q, \dots, Q\}$ and compact intensity domain $\lambda_t \in K \subset \mathbb{R}_+^d$. Between jumps, $d\lambda_t = \beta(\mu - \lambda_t)dt$, and a mark- i jump maps λ to $\Pi_K(\lambda + \beta \Gamma e_i)$, where Π_K is a fixed Lipschitz truncation map into K . Suppose the quote action set and the rectangular ambiguity set $\mathcal{A}$ are compact, all rewards, terminal payoffs, and controlled mark intensities are bounded and continuous, the rates and jump maps are uniformly Lipschitz on each inventory slice, $\sup_{\Gamma \in \mathcal{A}} \rho(\Gamma) < 1$, and the compact PDMP game satisfies the standard dynamic-programming and bounded-comparison principles for the rectangular admissible class, using standard PDMP, stochastic-control, and viscosity-solution machinery [20, 19, 18]. Then the robust market-making game has a bounded uniformly continuous value function, and it is the unique bounded viscosity solution of*

$$-\partial_t V = \sup_u \inf_{\Gamma \in \mathcal{A}} \left\{ r(q, \lambda, u, \Gamma) + \mathcal{L}^{u,\Gamma} V(q, \lambda) \right\}, \quad V(T, q, \lambda) = g(q, \lambda),$$

where $\mathcal{L}^{u,\Gamma}$ is the Hawkes PDMP generator. If the value is classically differentiable and the Hamiltonian has measurable saddle selectors, those selectors verify the robust value. Consequently, the truncated continuous-intensity game admits the claimed HJBI characterization under the stated compact-PDMP hypotheses. Passing to the global, untruncated near-critical model still requires the stability and comparison arguments developed next.

Theorem 4 (Conditional untruncated Hawkes Lyapunov bridge). *For the unprojected Hawkes intensity state $\lambda_t \in \mathbb{R}_+^d$, suppose there exist $v > 0$ and $\kappa > 0$ such that $\Gamma^\top v \leq (1 - \kappa)v$ for*

every $\Gamma \in \mathcal{A}$, fix $p > 1$, and suppose controlled mark rates satisfy $0 \leq \Lambda_i(q, \lambda, u, \Gamma) \leq \lambda_i$. Then $W_v(\lambda) = 1 + v^\top \lambda$ satisfies a uniform Foster–Lyapunov drift,

$$\mathcal{L}^{u, \Gamma} W_v(q, \lambda) \leq C_0 - C_1 W_v(\lambda),$$

and, after changing constants, the standard polynomial-weight version

$$\mathcal{L}^{u, \Gamma} W_v(\lambda)^p \leq C_{0,p} - C_{1,p} W_v(\lambda)^p.$$

Hence the untruncated controlled Hawkes PDMP is nonexplosive and has finite p th weighted maximal moments on finite horizons. For rewards and terminal payoffs with at most linear W_v -growth, compact truncations $\{W_v \leq R\}$ have exit probability at most $C_{T,p} W_v(\lambda)^p / R^p$, and the payoff tail error is of order R^{1-p} . Thus the compact projection is not needed for stability or tightness. What remains is uniqueness of the limiting HJBI on the unbounded intensity domain, addressed by the weighted comparison result below.

Theorem 5 (Sufficient weighted-comparison bridge for the untruncated HJBI). *Following the standard weighted-barrier and doubling-of-variables logic used in viscosity comparison arguments [18, 19], assume common Lyapunov subcriticality, compact ambiguity, and either a fixed compact control set or a compact control correspondence continuous in Hausdorff distance, continuous locally Lipschitz controlled rates with $0 \leq \Lambda_i \leq \lambda_i$, Lipschitz affine Hawkes jump maps, linearly growing rewards, and the weighted boundary condition*

$$\limsup_{R \rightarrow \infty} \sup_{\substack{t \in [0, T], q \in \{-Q, \dots, Q\} \\ W_v(\lambda) \geq R}} \frac{U(t, q, \lambda) - V(t, q, \lambda)}{W_v(\lambda)} \leq 0$$

for every subsolution U and supersolution V being compared. Then any upper-semicontinuous weighted-growth subsolution of the untruncated HJBI is dominated by any lower-semicontinuous weighted-growth supersolution with larger terminal data. Hence, within any W_v -weighted subclass with a fixed boundary condition at infinity, the untruncated robust Hawkes control value is the unique viscosity solution, and compact truncation solutions converge to it locally uniformly.

Theorem 6 (Interior robust quote optimizer sensitivity). *Fix the state and adjoint variables in the HJBI Hamiltonian, and let θ parameterize either the branching matrix or the center of the ambiguity set. Suppose that, near (δ_0, θ_0) , the robust reduced Hamiltonian $\mathfrak{H}(\delta, \theta) = \inf_{\Gamma' \in \mathcal{A}(\theta)} \mathcal{H}(\delta, \Gamma')$ has a unique C^1 active worst-case selector, the quote constraints are inactive, $\mathfrak{H}$ is C^2 in δ with continuous mixed derivative $D_{\delta\theta}^2 \mathfrak{H}$, δ_0 is the unique local maximizer at θ_0 , and $D_{\delta\delta}^2 \mathfrak{H} \preceq -mI$ locally. Then, for θ near θ_0 , there is a unique interior quote maximizer $\delta^*(\theta)$ in the corresponding quote neighborhood, and the map $\theta \mapsto \delta^*(\theta)$ is locally C^1 with*

$$D_\theta \delta^*(\theta_0) = - \left[D_{\delta\delta}^2 \mathfrak{H}(\delta_0, \theta_0) \right]^{-1} D_{\delta\theta}^2 \mathfrak{H}(\delta_0, \theta_0).$$

In the one-dimensional half-spread case, if e is a Perron-adverse structural direction and the directional mixed derivative in direction e is nonnegative, then the robust interior half-spread widens to first order. No-quote boundaries, spread caps, and changes in the active adversary fall outside this smooth interior calculation and instead require active-set analysis.

The computations use vectorized quote-grid evaluation and scenario-level parallelism. To make policy comparisons attributable to the policy rather than simulation noise, paired evaluations use fixed seeds and common random numbers.

6 Experiments

I first test the critical-scaling predictions in controlled ablations, then evaluate the robust quote policies in simulated and replayed order-flow environments. The experiments are ordered from mechanism-level checks to market-data diagnostics.

6.1 Scaling Ablations

Object	Family	Mean slope
Relative-slack premium	scalar	-2.000
Absolute-Gamma derivative	scalar	-3.000
Matrix resolvent	rank1	-3.191
Matrix resolvent	block	-3.213
Matrix resolvent	near-degenerate	-3.202
Matrix resolvent	sparse	-3.283

Table 2: Critical exponent ablation. Relative critical-slack ambiguity and absolute branching-matrix ambiguity produce the distinct blow-up rates predicted by the reduced-form theory.

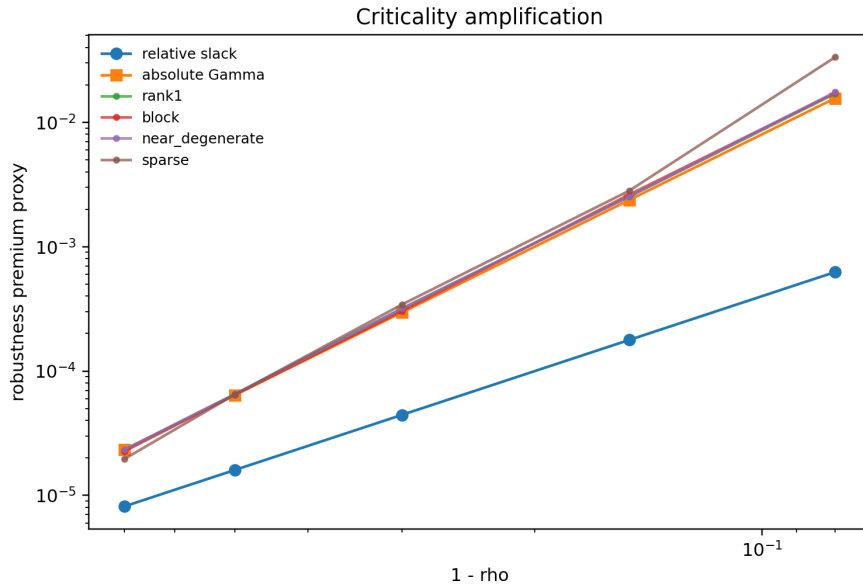


Figure 1: Criticality amplification under relative-slack ambiguity, absolute- Γ perturbations, and matrix-resolvent stress designs.

To separate the exponent from the particular stress design, I use a targeted spectral-gap and visibility ablation based on a two-mode branching matrix with eigenvalues ρ and $1 - k(1 - \rho)$ for $k \in \{2, 4, 8\}$. When the payoff loading is Perron-visible and the ambiguity set is aligned with the Perron direction, the estimated slope is -2.00 , with median normalized premium ratio 2.8125. A weakly Perron-visible loading recovers the same exponent after visibility normalization. By contrast, a Perron orthogonal loading has no positive Perron-aligned premium. When the adversary is instead allowed to move the second eigenmode, the orthogonal loading again has slope -2.00 as that mode approaches criticality. For weakly visible loadings, however, the median normalized second-mode

coefficient falls from 2.639 at $k = 2$ to 0.0365 at $k = 8$. The lower-bound statement therefore requires both Perron visibility and an ambiguity set that activates the Perron direction; otherwise the leading coefficient can vanish or be transferred to another near-critical mode.

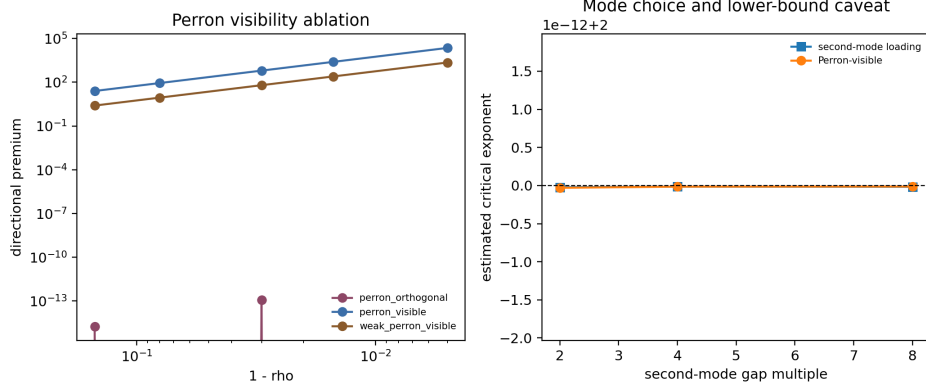


Figure 2: Spectral-gap and Perron-visibility ablation. Perron-visible loadings recover the square law under aligned ambiguity, while orthogonal loadings only respond to second-mode perturbations.

6.2 Policy Ablations

Policy	Certainty equivalent	CVaR5	Mean diff vs nominal	Paired 95% CI
robust_gamma	-13.944	-38.073	0.012	[0.005, 0.020]
robust_vol_only	-13.956	-38.087	0.000	[0.000, 0.000]
nominal_hawkes	-13.956	-38.087	baseline	—
robust_gamma_abs	-13.624	-37.321	0.306	[0.257, 0.359]
known_true_gamma_no_ambiguity	-13.952	-38.082	0.004	[-0.002, 0.011]
as_poisson	-13.974	-38.113	-0.017	[-0.026, -0.008]
liquidity_guard	-13.624	-37.321	0.306	[0.256, 0.355]

Table 3: Common-random-number policy ablation.

The intervals are paired bootstrap intervals over common-random-number paths, so each policy is compared with nominal on the same simulated realizations. In the main ablation, the absolute-Gamma and liquidity-guard policies have the highest reported certainty equivalents, CVaR5 values, and paired mean-wealth differences. The relative-slack robust-Gamma policy is closer to nominal, with a smaller but positive paired mean-wealth difference. I do not interpret the known-true- Γ benchmark as an oracle optimum: it is given the stressed ρ , but it does not carry an ambiguity premium, so it under-hedges model risk in this stress design. The table is best read as evidence that the policy response depends materially on the ambiguity normalization and on the withdrawal rule.

6.3 Policy Time-Step Audit

The simulator validation below shows that near-critical Hawkes counts can be sensitive to bin width, so I repeat the focused policy comparison at $\Delta t \in \{0.04, 0.02, 0.01\}$ for the stressed $\epsilon = 0.02$, $\hat{\rho} = 0.92$ scenario. Across the tested time steps, the relative-slack robust-Gamma policy keeps a positive mean-wealth advantage over nominal Hawkes: 0.042–0.060. The absolute-Gamma policy produces much larger mean-wealth gains, 1.154–1.366, but with negative lower paired 5% quantiles,

consistent with the broader point that ambiguity normalization changes the economic policy, not only the scale of the premium.

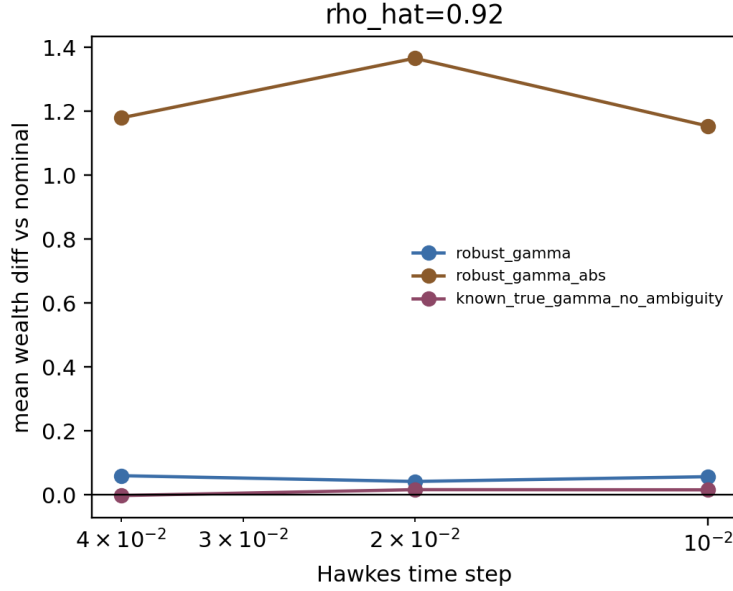


Figure 3: Focused policy time-step check. Positive values mean the policy outperforms nominal Hawkes on the same scenario.

6.4 Ogata Arrival Audit

I next compare the fast binned Hawkes recursion with event times generated by Ogata thinning and then binned onto the same decision grid. In the $\epsilon = 0.02$, $\hat{\rho} = 0.92$ check, the discrete recursion produces more events than the Ogata reference: mean total events are 226.8 under the discrete recursion and 185.5 under Ogata-binned arrivals. The relative-slack robust-Gamma policy is positive relative to nominal under discrete arrivals, with mean-wealth advantage 0.030, but is essentially flat and slightly negative under Ogata-binned arrivals, with mean-wealth difference -0.001 and paired 5/95 quantiles 0.002 and 0.049, reflecting a small number of negative tail paths. Absolute-Gamma is positive under both generators, with advantages 0.705 and 0.690 and paired 5/95 quantile ranges roughly $[-0.527, 2.651]$ and $[-0.524, 2.598]$. This check weakens any blanket claim that robust-Gamma dominates under every arrival generator, and it shows that the relative-slack result is generator-sensitive even though the absolute-Gamma result is similar across the two generators.

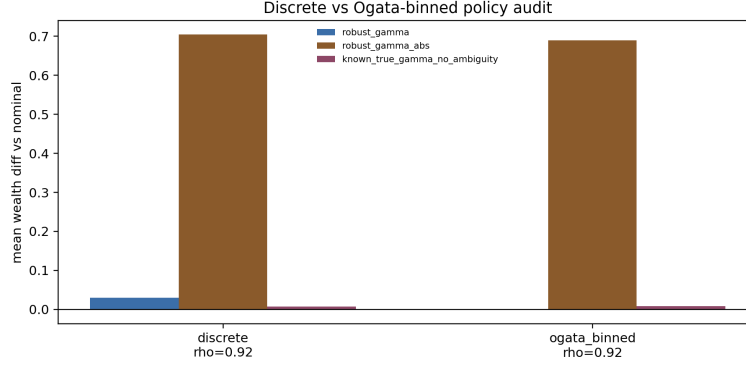


Figure 4: Focused policy check under fast discrete arrivals and Ogata-binned event arrivals for $\hat{\rho} = 0.92$.

6.5 Event-Queue Backtest

Finally, I run a reduced event-driven queue backtest on Ogata Hawkes event times. The evaluator holds the event path fixed across policies, updates quotes every $\Delta t = 0.02$, tracks a simple queue-ahead state, and records the side-time spent not quoting. For $\hat{\rho} = 0.92$ over 24 paths, robust-Gamma is nearly flat but positive relative to nominal, with mean-wealth difference 0.0125 and paired 5/95 quantiles $[0.004, 0.025]$, while absolute-Gamma and the liquidity guard each improve mean wealth by 0.342 with paired 5/95 quantiles $[0.115, 0.687]$. In this reduced run, fill rates are identical across policies at 0.943, the side-time no-quote fraction is 0.064, and the full no-quote time fraction is zero. This is not a production queue simulator, but it does provide an event-time execution stress test for the checked regime.

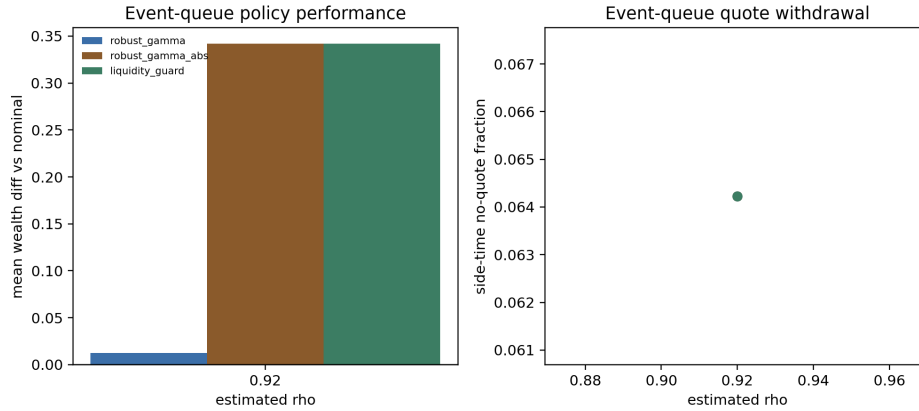


Figure 5: Reduced event-queue performance and shared side-time no-quote fraction.

6.6 Public LOBSTER Top-Of-Book Replay

The first replay asks whether the policy regimes observed in simulation also appear when replayed on public LOBSTER message/order-book streams. The replay uses up to the first 80,000 synchronized messages per ticker, posts active policies at the displayed L1 quotes subject to an inventory cap, depletes a simple queue-ahead proxy using observed execution sizes, and marks to the observed midprice. Under the calibrated side-aware spectral radii, all tested policies remain inside the L1 activity threshold, so the replay does not force an artificial separation: mean terminal wealth is 341.691, mean fills are 3071.8, and side-time no-quote share is 0.198 for each policy. Under the

near-critical stress setting $\hat{\rho} = 0.97$, nominal and relative-slack robust-Gamma remain active on the same real event streams, while absolute-Gamma and the liquidity guard fully withdraw: their mean wealth difference versus nominal is -341.691 , mean fills are zero, and side-time no-quote share is one. This should not be read as a production queue simulator. Its narrower point is that the quote-withdrawal transition appears in this public replay when the same policies are placed on observed L1 event paths and prices.

The next replay keeps the L1 restriction but makes quote placement visible. Each continuous policy offset is rounded into one of four displayed states: join the best quote, improve inside the spread, rest away from L1, or withdraw. Away quotes are not filled, because one-level data cannot identify hidden depth. Under calibrated side-aware spectral radii, tick rounding still makes policy PnL values coincide, but the state mix explains where the quotes would have gone: mean side-time is 0.024 joining, 0.428 improving, 0.448 away from L1, and 0.101 withdrawn. Under $\hat{\rho} = 0.97$, absolute-Gamma and the liquidity guard fully withdraw. This is a more informative public-data check than the binary replay, while still stopping short of production execution claims.

I then let away quotes interact with displayed depth by replaying the uniform level-10 LOBSTER panel. If a rounded quote rests inside the displayed ten-level ladder, the replay initializes queue-ahead from all better displayed levels plus a same-level queue fraction; observed executions and visible depth changes can then deplete that queue. Under calibrated side-aware spectral radii, policies still coincide after tick rounding, but the state mix is more informative: mean side-time is 0.018 joining L1, 0.428 improving, 0.408 resting at visible depth, and 0.146 withdrawn, with mean visible depth rank 2.431 and mean fills 1420.2. Under $\hat{\rho} = 0.97$, absolute-Gamma and the liquidity guard fully withdraw, with mean wealth difference -120.216 versus nominal. The replay remains conservative about hidden liquidity and exact priority, but the test is no longer L1-only.

The priority-aware replay adds explicit priority accounting within the public level-10 book. At each quote reset, the synthetic order is placed behind the full displayed size at the same price. Later same-price limit orders are tracked by order identifier as behind the synthetic quote, so cancellations or executions of those later orders cannot improve its queue position. Queued fills are credited only from residual same-price execution volume after displayed queue-ahead has been depleted; exact queue exhaustion with no residual volume is recorded but not filled, and partial fills are credited by residual lots.

This stricter rule shows how much the earlier displayed-depth replay depended on a heuristic same-level queue cap. Under calibrated side-aware spectral radii, mean fill events are 21.6 but mean filled lots are only 15.53, mean terminal wealth is -7.839 , and mean side-time is 0.019 joining L1, 0.530 improving, 0.422 resting at visible depth, and 0.029 withdrawn. Under $\hat{\rho} = 0.97$, absolute-Gamma and liquidity-guard policies fully withdraw and avoid the nominal priority-replay loss, giving mean wealth difference $+5.964$ versus nominal. The regenerated artifact also records queue-position invariants: in the calibrated baseline, mean initial displayed queue ahead is 62.03 lots across 1810.4 visible quote resets, queued fills average 1.2, inside-spread improve fills average 20.4, partial-fill events average 7.6, the maximum zero-residual prevented-fill count is 1, and the maximum queue-violation count is zero. This is the most queue-position-aware replay in the package, but it still cannot observe hidden liquidity or recover order identifiers for anonymous queue ahead already present when the synthetic quote arrives.

As a depth supplement, I also replay the deepest public LOBSTER files available for the sample panel: level-50 AAPL and MSFT on their shorter one-hour public window. This supplement uses all 50 displayed levels in those files and records mean fill events 23.0, mean filled lots 16.82, mean queued fills 2.5, mean improve fills 20.5, mean partial-fill events 8.0, max visible depth rank 5, and maximum queue-violation count zero. Because only two tickers have public level-50 samples in the mirror, this is a supplement rather than the headline five-ticker panel.

I also run a priority-assumption sensitivity grid for the nominal policy. The grid varies the fraction of displayed same-price queue placed ahead of the synthetic order over $\{0, 0.5, 1\}$ and a queue-stress multiplier over $\{1, 1.5, 2\}$. The multiplier is applied to the public displayed queue-ahead proxy as a conservative stress test; it is not evidence that hidden depth has been observed. On the calibrated level-10 panel, the optimistic displayed priority setting $(0, 1)$ gives mean filled lots 16.53; the strict displayed-priority baseline $(1, 1)$ gives mean filled lots 15.53; and the most conservative tested queue-stress setting $(1, 2)$ gives mean filled lots 14.87. The narrow range is useful precisely because it is negative evidence: public displayed books rarely identify enough residual same-price execution volume to make hidden-priority assumptions strongly estimable. Across the sensitivity grid, the maximum queue-violation count remains zero.

Finally, I check whether the public messages support the queue analysis near the touch. I reconstruct the observable level-10 book from messages and compare it to the official LOBSTER snapshots. The reconstructor starts from the first official snapshot, tracks all post-start order IDs, and re-anchors every 100 messages so the check measures local reconstruction quality rather than all-day truncation drift. Across 399,995 processed events and 40,000 compared snapshots, panel mean top-of-book price match is 0.9997, full 10-level price match is 0.9376, top-of-book size MAE is 0.056 shares, and full-depth size MAE is 38.1 shares. These results are consistent with local near-touch priority analysis from public messages, while deeper queue-position claims should remain conservative.

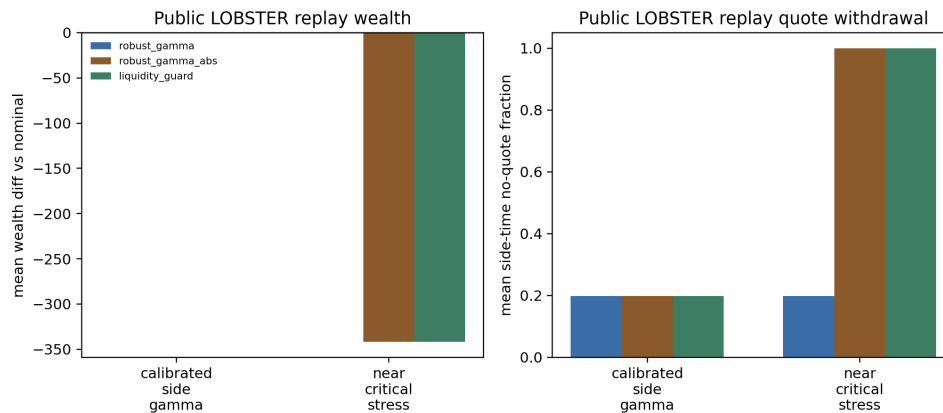


Figure 6: Public LOBSTER top-of-book replay on actual message/order-book streams. Calibrated policies coincide under the simple binary rule; in the near-critical stress case, absolute-Gamma ambiguity withdraws quotes on the same data paths.

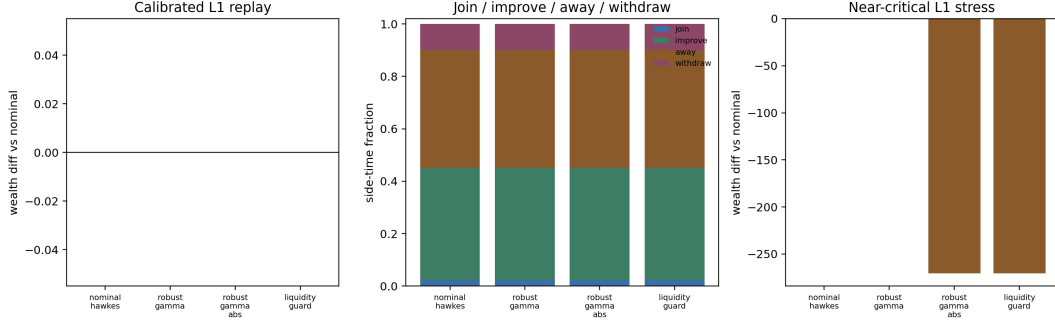


Figure 7: Offset-aware L1 quote replay. Tick rounding still makes calibrated policy wealth coincide, but the replay shows where quotes would be placed and when near-critical absolute-Gamma ambiguity withdraws them.

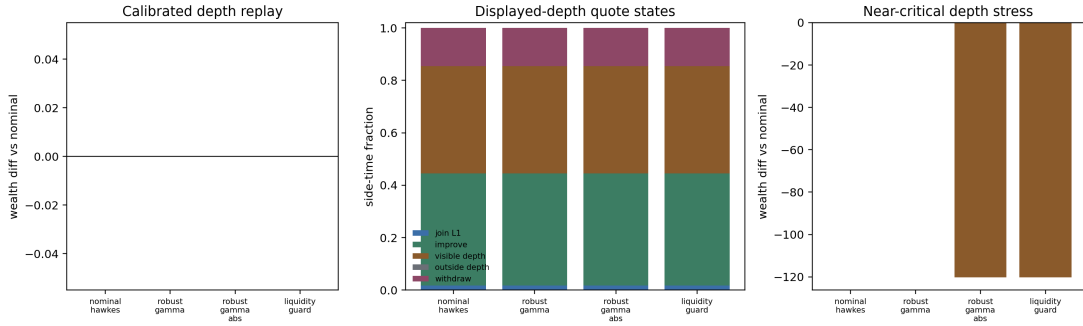


Figure 8: Displayed-depth level-10 LOBSTER quote replay. Away-from-L1 quotes can fill only when they rest inside the visible book and observed executions or depth changes deplete displayed queue ahead.

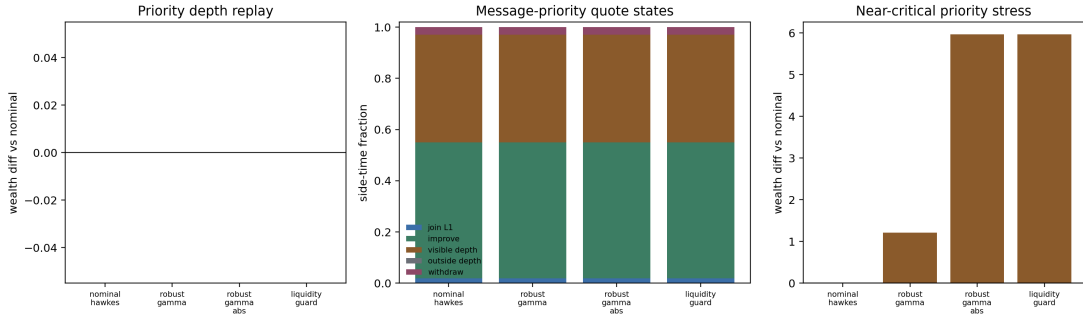


Figure 9: Residual-volume priority-aware level-10 LOBSTER quote replay. Same-price orders arriving after the synthetic quote are kept behind it; queued fills require residual execution volume after displayed queue-ahead depletion.

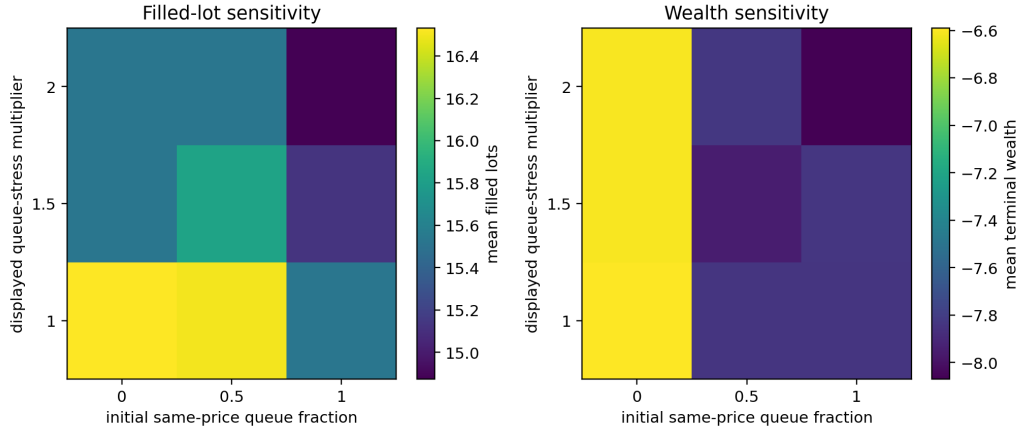


Figure 10: Priority-assumption sensitivity for the nominal level-10 replay. The grid varies same-price displayed priority and displayed queue stress; filled lots remain rare, showing the public-data limit of same-price priority identification.

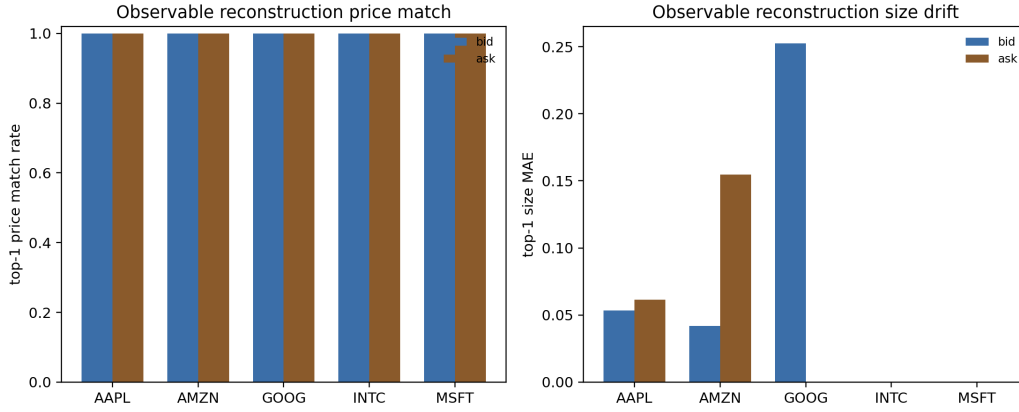


Figure 11: Observable order-book reconstruction check. Re-anchored message-level reconstruction matches top-of-book prices closely, while full-depth size drift remains visible.

6.7 Simulator Validation

The slow Ogata reference simulator makes the numerical issue visible: coarse time discretization can create spurious critical amplification. At $\rho = 0.90$, the relative mean-count bias falls from 3.07 at $\Delta t = 0.08$ to 0.03 at $\Delta t = 0.01$. At $\rho = 0.97$, even $\Delta t = 0.005$ leaves about 0.20 relative bias in the current short-horizon setting.

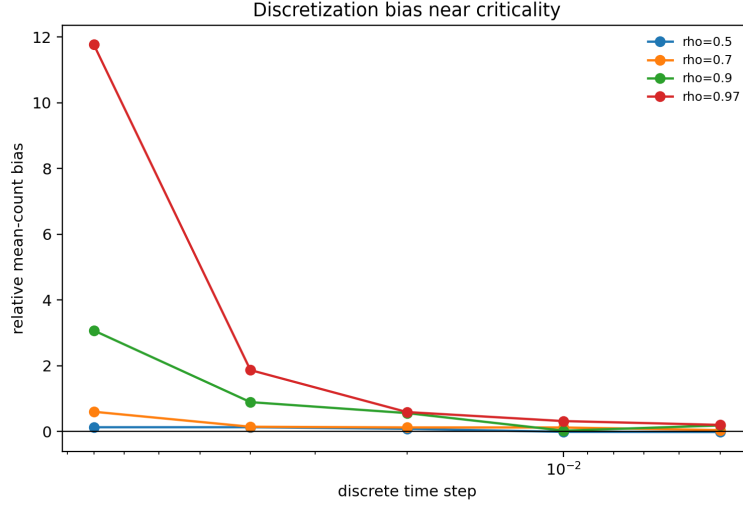


Figure 12: Time-discretization bias grows sharply near criticality.

6.8 Calibration Noise

A simple Fano-based criticality estimator points to the same practical concern: near-critical calibration is fragile. For $\rho = 0.97$, short horizons under-estimate criticality, while longer horizons can over-shoot because rare clusters dominate the count variance. This is why the empirical $\hat{\rho}$ estimates should be read with uncertainty, not as exact structural inputs.

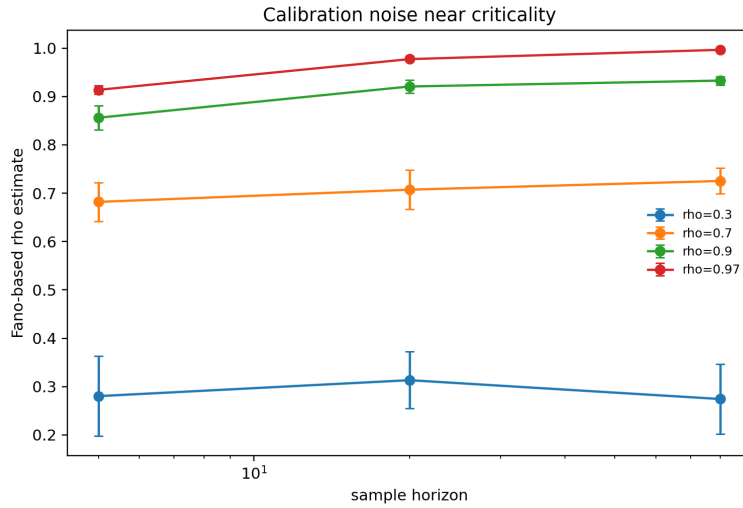


Figure 13: Fano-based criticality estimates across horizons.

6.9 Robust Dynamic Programming

The finite-scenario DP includes explicit bid-side and ask-side no-quote actions. In the reported reduced run, the raw DP action grid contains 4,044 two-sided quote states, 618 bid-only states, 618 ask-only states, and no full no-quote states. The first-step, zero-inventory rows remain two-sided through the tested grid, which reaches $\hat{\rho} = 0.92$. Averaged over the DP grid, no-quote rates are 22.3% under relative-slack ambiguity and 24.5% under absolute-Gamma ambiguity; full two-sided withdrawal rates are zero for both. This is still a reduced Bellman approximation, not a queue-calibrated quote-refusal simulator comparable one-for-one with [5].

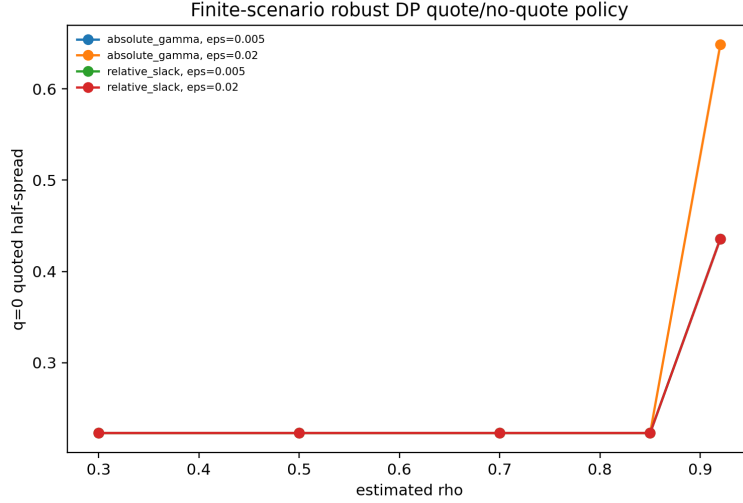


Figure 14: Reduced robust-DP half-spreads and no-quote switching as criticality increases.

I also run a quote-map sensitivity diagnostic linking the local optimizer theorem to the implemented policy map. For the smooth implemented half-spread, nominal and relative-Gamma robust policies recover critical exponent 3.000 for $d\delta/d\rho$, while absolute-Gamma robust policies approach exponent 4.000 in the uncapped interior region. Capped or no-quote points are marked as nonsmooth active-set cases rather than treated as classical derivatives.

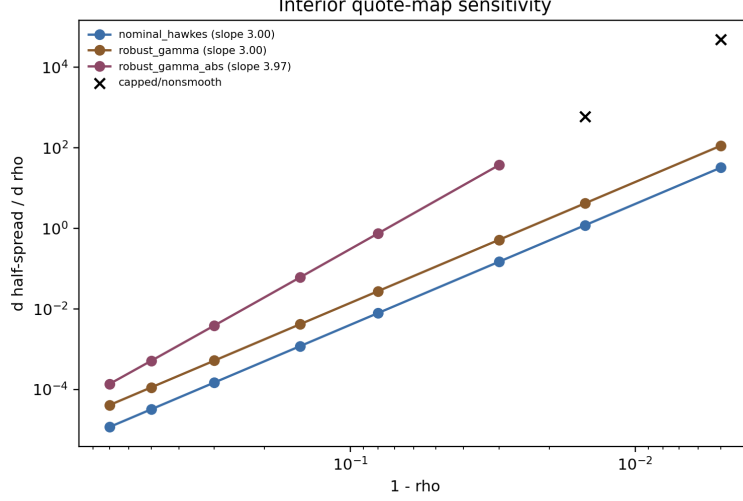


Figure 15: Interior quote-map sensitivity. Smooth relative-Gamma policies have third-order derivative blowup, while absolute-Gamma ambiguity adds one more critical derivative factor until spread caps bind.

6.10 Public Data Panels

The public datasets are not used to prove the structural ambiguity theorem, because the true Γ is unobserved. Instead, they document clustered order flow and overdispersed liquidity states in the data used for the diagnostics. Binance aggregate trades are fetched from the public archive at <https://data.binance.vision/?prefix=data/spot/daily/aggTrades/>.

Ticker	Rows	Events/sec	Event Fano	Mean spread
AAPL	118497	5.064	19.460	0.155
AMZN	57515	2.458	13.329	0.136
GOOG	49482	2.115	17.185	0.311
INTC	404986	17.307	220.650	0.013
MSFT	411409	17.582	196.485	0.013

Table 4: Public LOBSTER sample sanity panel.

Symbol	Agg. trades	Agg. trades/sec	Event Fano	ACF1
BTCUSDT	1364603	15.794	200.274	0.342
ETHUSDT	597417	6.915	29.730	0.241
SOLUSDT	218278	2.526	12.240	0.201

Table 5: Public Binance aggregate-trade event panel for 2024-01-15.

Symbol	Rows	Rel. spread bps	Return std	Depth Fano
BTC	10081	0.163	0.000687	33.989
ETH	10081	0.591	0.000910	513.703
SOL	10080	0.911	0.001110	1697.605

Table 6: Public crypto L2 depth sanity panel.

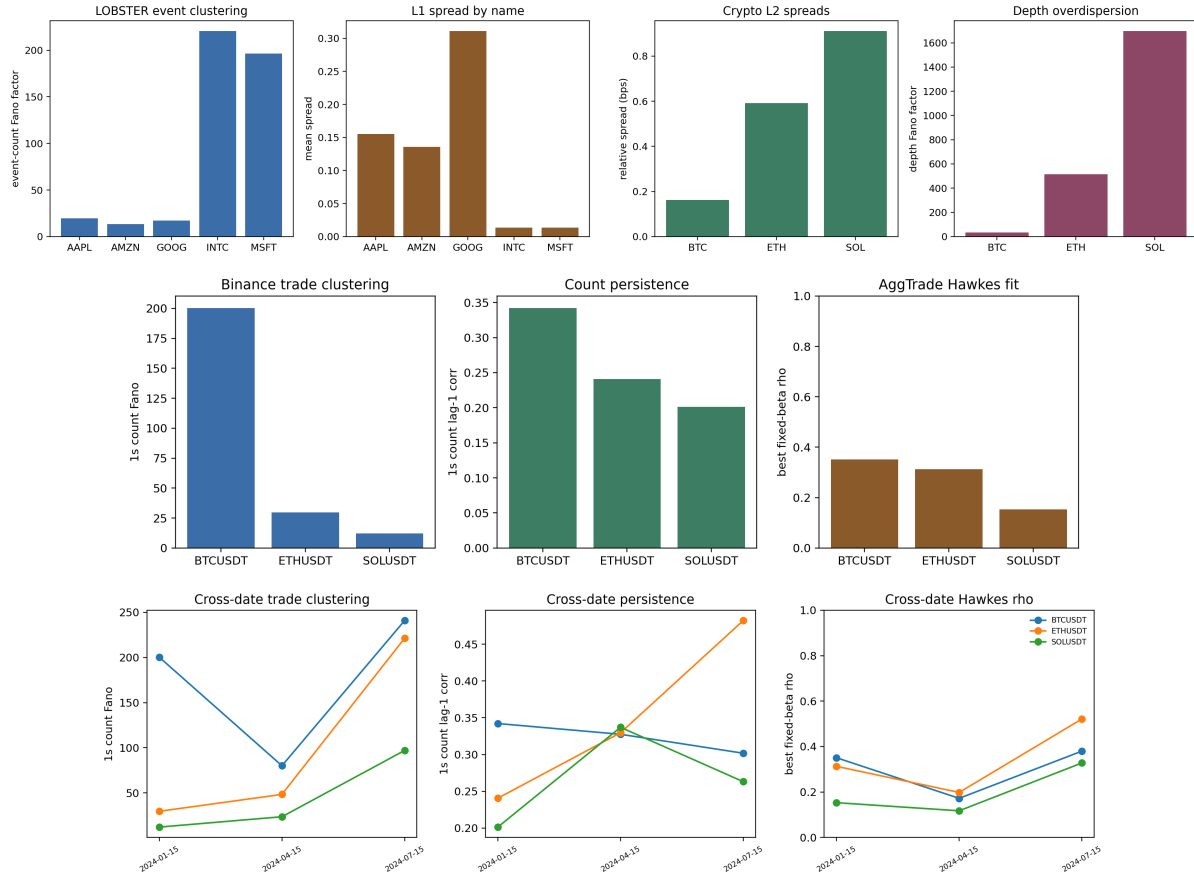


Figure 16: Public-data sanity figures: LOBSTER event clustering and spreads, crypto L2 spreads and depth overdispersion, Binance aggregate-trade clustering, fixed-beta Hawkes diagnostics, and cross-date robustness.

6.11 Corrected Hawkes Calibration Diagnostics

As a check before interpreting real-data Hawkes fits, I validate the corrected single-exponential MLE on Ogata-simulated paths with known parameters. The validation recovers moderate and near-critical branching ratios without beta-bound failures.

ρ true	β true	ρ MAE	β MAPE	Beta cap rate
0.30	2.0	0.043	0.201	0.000
0.70	4.0	0.019	0.026	0.000
0.90	8.0	0.016	0.039	0.000

Table 7: Synthetic Ogata validation for the corrected Hawkes MLE.

I also validate the fixed-beta marked multivariate estimator on three-type Ogata Hawkes paths with known branching matrices. For target spectral radii 0.35 and 0.65 at $\beta = 3$, the marked estimator has spectral-radius MAE 0.025 and 0.023, relative Frobenius branching-matrix error 0.234 and 0.133, and a success rate of 1.000. This check is important because the public-data marked calibrations below are only meaningful if the estimator can recover a known multitype Γ under controlled simulation.

ρ true	Mean events	ρ MAE	$\ \hat{\Gamma} - \Gamma\ _F / \ \Gamma\ _F$	Success
0.35	1209.8	0.025	0.234	1.000
0.65	2171.2	0.023	0.133	1.000

Table 8: Synthetic Ogata validation for the fixed-beta marked multivariate Hawkes estimator.

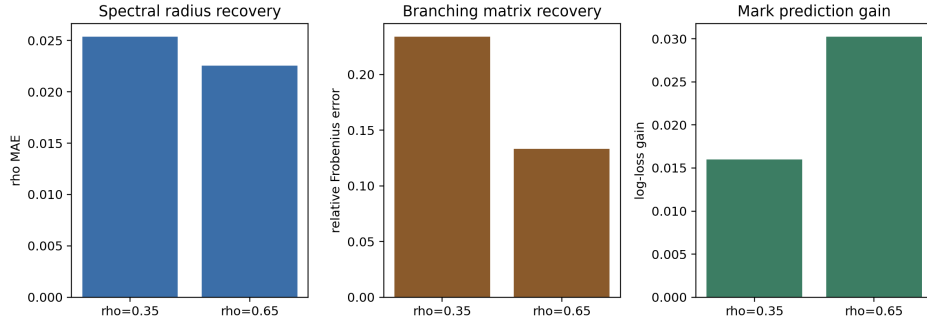


Figure 17: Marked multivariate Ogata validation: spectral-radius recovery, branching-matrix recovery, and conditional mark-prediction gain.

The Binance aggregate-trade panel provides a separate public check on crypto event clustering. On 2024-01-15, BTCUSDT, ETHUSDT, and SOLUSDT have 218k to 1.36m aggregate trades, one-second event-count Fano factors from 12.24 to 200.27, and lag-one count autocorrelations from 0.201 to 0.342. Fixed beta Hawkes fits over all aggregate trades, buy-aggressor trades, and sell-aggressor trades select $\beta = 100$ in the current grid. All-trade branching ratios are 0.351 for BTCUSDT, 0.313 for ETHUSDT, and 0.153 for SOLUSDT; sell-aggressor branching is higher for BTCUSDT and ETHUSDT, at 0.493 and 0.463. This is useful cross-asset evidence of clustered event flow, but aggregate trades do not include order placement, cancellation, or queue state, so the panel should not be read as an execution benchmark.

To avoid relying on one crypto day, I repeat the Binance check on 2024-01-15, 2024-04-15, and 2024-07-15. Across all-event fixed-beta fits, branching-ratio ranges are 0.172–0.380 for BTCUSDT,

0.198–0.521 for ETHUSD, and 0.117–0.328 for SOLUSD. The median all-event branching ratios remain 0.351, 0.313, and 0.153, respectively.

On raw LOBSTER message arrivals, the all-message single-exponential fit remains a misspecification diagnostic rather than a structural calibration target. The corrected all-message panel still places β at the upper bound for every ticker under the 100k-event cap, with fitted branching ratios between 0.64 and 0.76. Event-type splits and fixed-beta profiles show that branching estimates are sensitive to the assumed decay scale; timestamp aggregation sharply reduces the fitted decay, with median β falling from 148.4 on raw events to 23.4 at 1ms aggregation and 1.46 at 10ms aggregation. A fixed-beta two-scale diagnostic adds a multiscale check but does not remove the misspecification warning: among the tested pairs, every ticker/event-group selects $(\beta_{\text{slow}}, \beta_{\text{fast}}) = (1, 100)$, and median fast-excitation shares are 0.764 for all messages, 0.723 for limits, 0.660 for cancel/delete events, and 0.871 for executions. Execution events remain the hardest to fit, with median residual KS statistic about 0.415. This supports the methodological point that raw unmarked message flows mix multiple clocks and packet-level bursts, so, in this setting, robust market-making experiments should either use marked or multiscale calibration diagnostics, or report residual diagnostics [1, 17].

The package therefore also includes a grouped marked multivariate Hawkes fit on limit, cancel/delete, and execution event groups. With fixed exponential decay, AIC selects $\beta = 100$ for all five public LOBSTER tickers. The estimated branching matrices are stable, with spectral radii from 0.367 to 0.622, and the conditional mark log-loss improves over unconditional event-frequency prediction by 0.005 to 0.114 nats per event. The median branching matrix has the strongest source column from execution events, with median total outgoing branching about 0.923, and the largest median target row into limit intensity, about 0.897. Residual KS statistics still range from 0.050 to 0.217, so the fit should be read as a reproducible marked calibration diagnostic, not as a production-grade structural Γ estimate.

Finally, to align the empirical object with the bid/ask control model, I fit a six-mark side-aware Hawkes model using LOBSTER direction labels: limit, cancel/delete, and execution events on buy and sell sides. AIC again selects $\beta = 100$ for all five tickers. The full six-mark branching matrices remain stable, with spectral radii from 0.369 to 0.624, while conditional six-mark log-loss improves over unconditional mark frequencies by 0.101 to 0.292 nats per event. Median side-aggregated branching is strongest within side, about 1.354 from buy to buy and 1.403 from sell to sell, with smaller cross-side totals of about 0.492 from buy to sell and 0.414 from sell to buy. The aggregate row sums can exceed one because they sum many target/source entries; stability is assessed by the spectral radius of the full six-mark matrix.

I then condition the six-mark time-rescaling residuals on observable L1 state variables: event group, side, within-ticker size bucket, spread bucket, top-depth bucket, and order-book imbalance bucket. This is a misspecification check rather than another fitted model. Across the five public LOBSTER samples, wide-spread states have median residual mean 0.886 and median KS statistic 0.220, while tight-spread states have median residual mean 1.110 and median KS statistic 0.092. Large-size events have residual mean 0.828 versus 1.018 for small events. Execution marks show the largest conditional mark-prediction gain, with median log-loss improvement 0.384 nats per event. The result sharpens the empirical conclusion: side-aware excitation is informative, but the residuals still depend materially on size and book state.

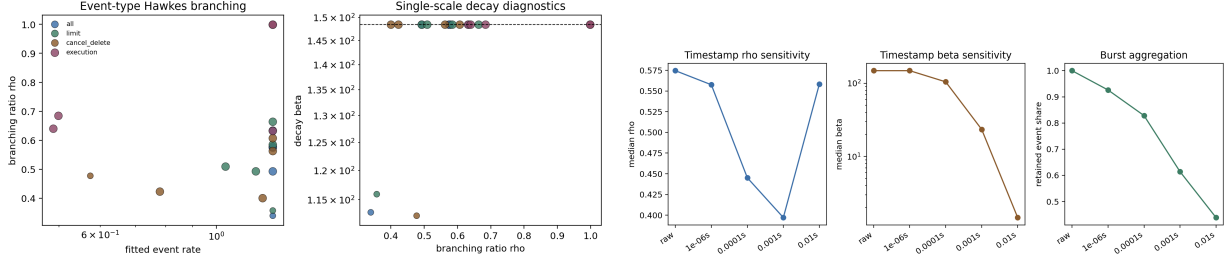


Figure 18: Corrected public-data Hawkes diagnostics: event-type branching/decay and timestamp-resolution sensitivity.

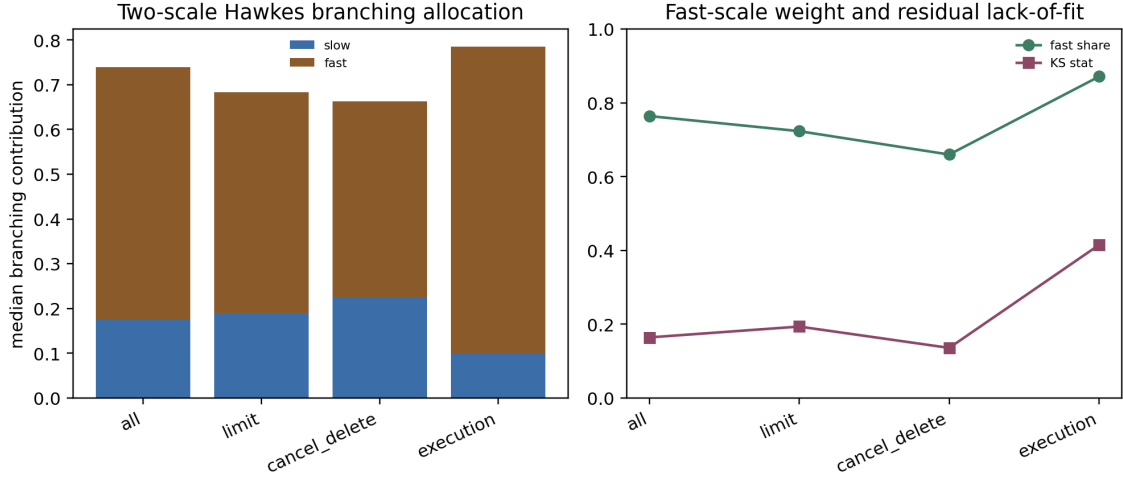


Figure 19: Fixed-beta two-scale public-data Hawkes diagnostics. The selected fits concentrate excitation on the fast component, especially for execution events, while residual tests still flag misspecification.

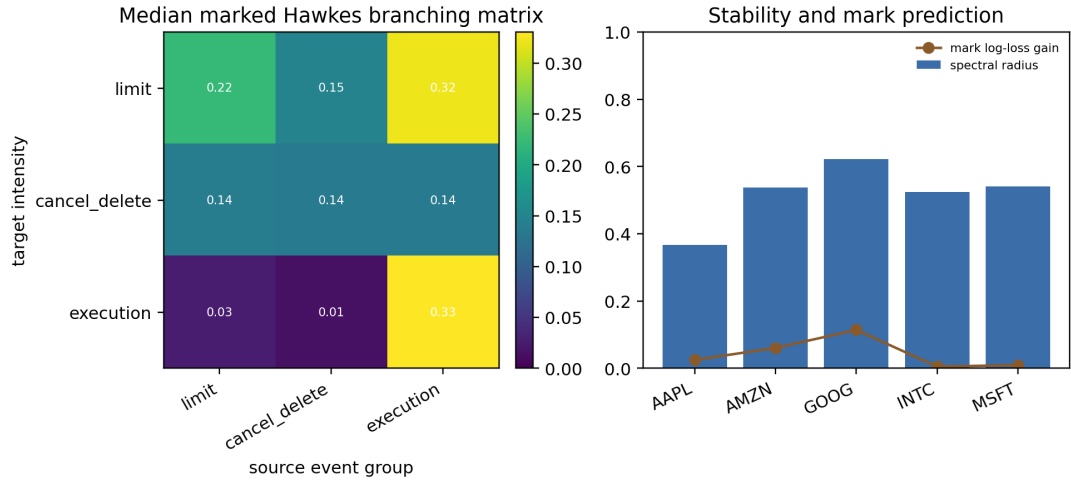


Figure 20: Grouped marked multivariate Hawkes diagnostics on LOBSTER limit, cancel/delete, and execution events. The fixed-beta fits estimate stable branching matrices and improve conditional mark prediction, but residual tests still point toward richer side, size, and multiscale calibration.

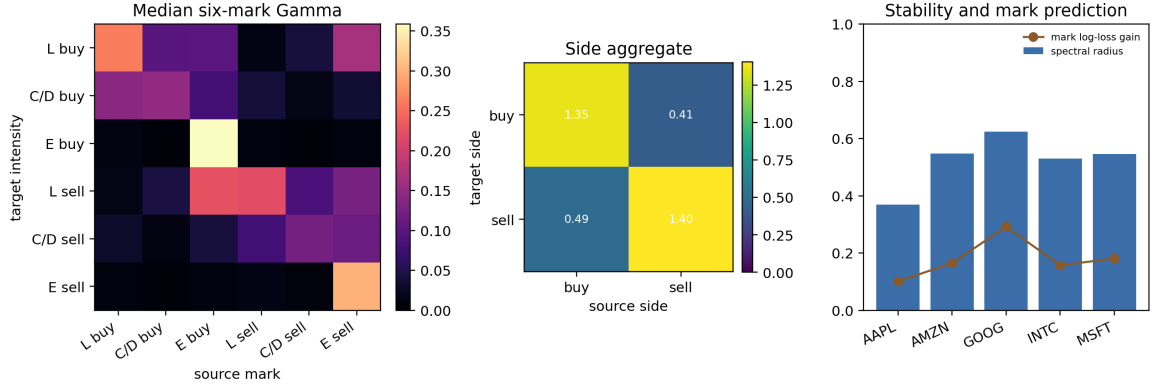


Figure 21: Side-aware marked multivariate Hawkes diagnostics using LOBSTER direction labels. The six-mark fits bring the empirical calibration object closer to bid/ask robust control, while remaining fixed-decay and message-side rather than queue-position or size-aware.

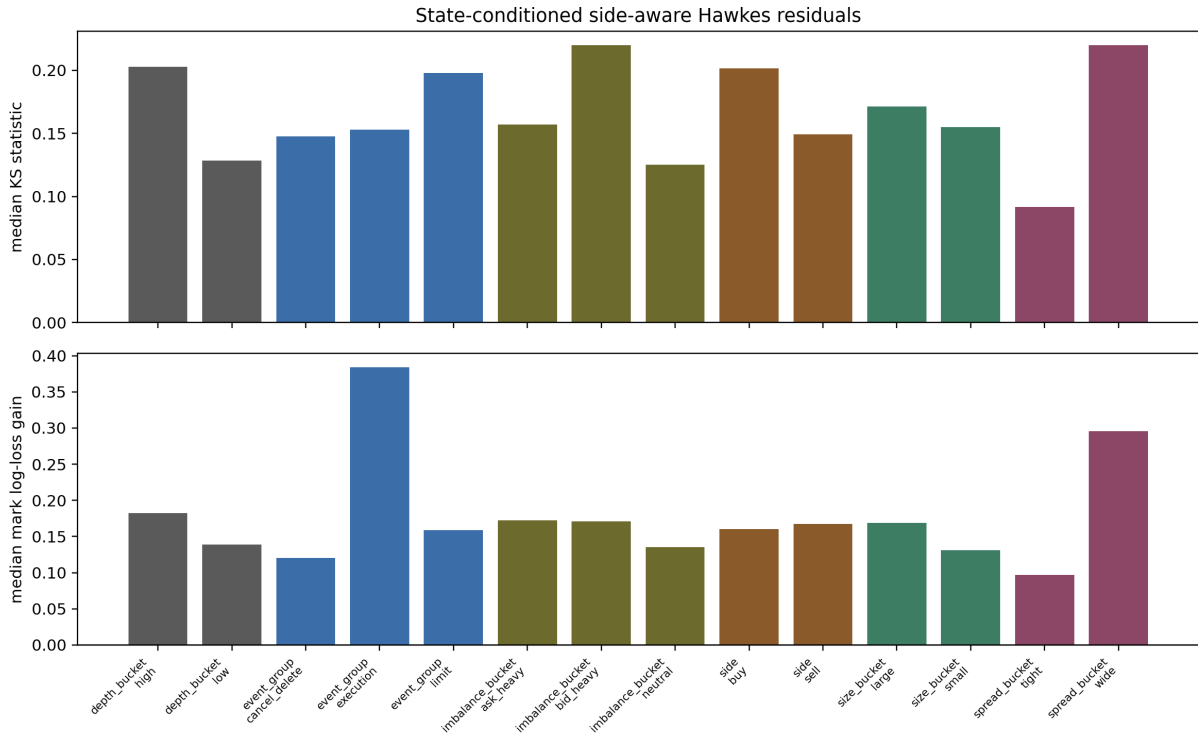


Figure 22: State-conditioned residual diagnostics for the side-aware six-mark Hawkes fit. Residual deviations and mark-prediction gains vary by spread, depth, imbalance, and event-size bucket, so the public-data calibration is best read as a diagnostic input rather than a complete structural law.

7 Limitations and Scope Conditions

This paper develops a conditional multitype HJBI control link for the stated exponential Hawkes model class. The argument combines compact truncated HJBI verification under compact-PDMP DPP/comparison hypotheses, untruncated Lyapunov stability with $p > 1$ tail control, pairwise weighted comparison and uniqueness in fixed-boundary W_v -weighted subclasses, local differentiability of smooth interior quote optimizers, and a consolidated argument assembling these pieces. It does not extend automatically to arbitrary state-dependent, learned, multiscale, or power-law Hawkes kernels without rechecking the stated Lyapunov and comparison assumptions. No-quote and active-adversary switches are treated as nonsmooth control boundaries, not as classical optimizer derivatives.

The top-of-book, L1, displayed-depth, residual-volume priority-aware level-10, deepest-public level-50, priority-sensitivity, and observable-reconstruction LOBSTER checks should be read as public-data execution checks rather than production execution claims, because hidden liquidity and anonymous initial queue priority remain unobserved. Binance aggregate trades should be read as event-clustering evidence rather than a queue-calibration source. These scope conditions delimit the paper’s claim: the reported results concern structural Hawkes ambiguity, criticality amplification, and calibration fragility in the stated Hawkes market-making experiments.

References

- [1] E. Bacry, I. Mastromatteo, and J.-F. Muzy. Hawkes processes in finance. arXiv:1502.04592, 2015.
- [2] B. Law and F. Viens. Market making under a weakly consistent limit order book model. arXiv:1903.07222, 2019.
- [3] P. Kumar. Deep Hawkes Process for High-Frequency Market Making. Journal of Banking and Financial Technology, 2024. arXiv:2109.15110, doi:10.1007/s42786-024-00049-8.
- [4] D. Raffaelli, R. G. Cestari, D. Marazzina, and S. Formentin. Forecasting Bitcoin price movements using multivariate Hawkes processes and limit order book data. Decisions in Economics and Finance, 2026. doi:10.1007/s10203-026-00570-z.
- [5] Z. Wang, C. Ventre, and M. Polukarov. Robust Market Making: To Quote, or not To Quote. In ICAIF 2023: Proceedings of the Fourth ACM International Conference on AI in Finance, pages 664–672, 2023. doi:10.1145/3604237.3626858. arXiv:2508.16588, 2025.
- [6] Z. Wang, C. Ventre, and M. Polukarov. ARL-Based Multi-Action Market Making with Hawkes Processes and Variable Volatility. In Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF 2024), pages 437–444, 2024. doi:10.1145/3677052.3698695. arXiv:2508.16589, 2025.
- [7] L. Lalor and A. Swishchuk. Event-Based Limit Order Book Simulation under a Neural Hawkes Process: Application in Market-Making. arXiv:2502.17417, 2025.
- [8] H. Guo, J. Lin, and F. Huang. Market Making with Deep Reinforcement Learning from Limit Order Books. In 2023 International Joint Conference on Neural Networks (IJCNN), 2023. arXiv:2305.15821, doi:10.1109/IJCNN54540.2023.10191123.

- [9] J. Jiang, C. Yang, X. Wang, Z. Li, X. Huang, and B. Li. Resolving Latency and Inventory Risk in Market Making with Reinforcement Learning. arXiv:2505.12465, 2025.
- [10] K. Jain, N. Firoozye, J. Kochems, and P. Treleven. An Impulse Control Approach to Market Making in a Hawkes LOB Market. arXiv:2510.26438, 2025.
- [11] S. El Karmi. A Deterministic Limit Order Book Simulator with Hawkes-Driven Order Flow. arXiv:2510.08085, 2025.
- [12] P. Noble, M. Rosenbaum, and S. Souilmi. Bridging the Reality Gap in Limit Order Book Simulation. arXiv:2603.24137, 2026.
- [13] A. Kimura. Extended State-dependent Hawkes Process for Limit Order Books: Mathematical Foundation and the Reproduction of Volatility Signature Plots. arXiv:2604.23961, 2026.
- [14] G. Szymanski and W. Xu. Mean-Field Limits for Nearly Unstable Hawkes Processes. arXiv:2501.11648, 2025.
- [15] S. El Karmi. Scaling Limits of Bivariate Nearly-Unstable Hawkes Processes and Applications to Rough Volatility. arXiv:2605.03703, 2026.
- [16] S. J. Hardiman, N. Bercot, and J.-P. Bouchaud. Critical reflexivity in financial markets: a Hawkes process analysis. arXiv:1302.1405, 2013.
- [17] Y. Ogata. Statistical models for earthquake occurrences and residual analysis for point processes. *Journal of the American Statistical Association*, 83(401):9–27, 1988.
<https://doi.org/10.1080/01621459.1988.10478560>.
- [18] M. G. Crandall, H. Ishii, and P.-L. Lions. User’s guide to viscosity solutions of second order partial differential equations. *Bulletin of the American Mathematical Society*, 27(1):1–67, 1992. doi:10.1090/S0273-0979-1992-00266-5.
- [19] W. H. Fleming and H. M. Soner. *Controlled Markov Processes and Viscosity Solutions*. 2nd ed., Springer, 2006. doi:10.1007/0-387-31071-1.
- [20] M. H. A. Davis. *Markov Models and Optimization*. Chapman & Hall, 1993. ISBN 9780412314100.

Mathematical Appendix for Structural Hawkes Ambiguity in Near-Critical Market Making

1 Purpose

This companion appendix records the proof details that would slow the main paper’s narrative. MESA means market-making excitation-structure ambiguity. The first sections cover the scalar amplification result, a Perron-visible multitype bridge, the finite-state Bellman layer used in the dynamic program, and a compact HJBI verification result for the truncated multitype game. The later sections spell out the extra conditions needed beyond truncation: robust subcriticality, weighted comparison, and local sensitivity of interior quote optimizers. The scope is deliberately narrow: the appendix covers only the model class used in the main paper and makes the assumptions needed for extensions explicit.

2 Notation Used Below

Symbol	Domain	Meaning
ρ	$[0, 1)$	Scalar branching ratio or Perron root $\rho(\Gamma)$, depending on context.
Γ	$\mathbb{R}_+^{d \times d}$	Multitype Hawkes branching matrix.
μ, λ	$\mathbb{R}_+^d$	Baseline intensity and current intensity vector.
r, ℓ	$\mathbb{R}_+^d$	Right and left Perron vectors, normalized by $\ell^\top r = 1$.
$E, E_{\text{rel}}, E_{\text{abs}}$	Matrix perturbations	Generic, relative-slack, and absolute branching-matrix perturbations.
$\varepsilon, \varepsilon_{\text{rel}}$	positive scalars	Absolute and relative ambiguity radii.
$V_{\text{risk}}, V_d, V_{\text{bi}}$	real-valued functions	Reduced risk proxies used to isolate the resolvent singularity.
q	$\{-Q, \dots, Q\}$	Inventory state in the Bellman and HJBI games.
K	compact subset of $\mathbb{R}_+^d$	Truncated continuous-intensity domain.
$U, \mathcal{A}$	compact or finite sets	Quote/no-quote controls and rectangular ambiguity set.
$L^{u, \Gamma}, \mathcal{L}^{u, \Gamma}$	generators	Finite-state and continuous-intensity controlled Hawkes generators.
W_v, κ	Lyapunov function and drift margin	Weighted intensity Lyapunov function and common subcriticality margin.
J_i, K_R	jump map and compact set	Untruncated Hawkes mark- i jump map and Lyapunov sublevel truncation set $\{W_v \leq R\}$.
U, V, F	weighted viscosity functions	Generic subsolution, supersolution, and solution candidate in the comparison theorem.
δ, θ	quote vector and structural parameter	Local quote action and finite-dimensional branching/ambiguity parameter in the optimizer sensitivity theorem.
$\mathfrak{H}, \delta^*$	reduced Hamiltonian and optimizer	Robust local Hamiltonian after Nature's minimization and its interior quote maximizer.
T, g, r	finite horizon, payoff, reward	Terminal time, terminal payoff, and running reward.

Unless a theorem says otherwise, asymptotic constants are taken uniformly over the compact regularity classes stated there. The same letters may denote a scalar object in the scalar section and its multitype analogue later; each section restates the needed domain.

3 Scalar Hawkes Setup

Let N_t be a stationary linear Hawkes process with exponential kernel

$$\lambda_t = \mu + \int_0^{t^-} \beta \rho e^{-\beta(t-s)} dN_s, \quad 0 \leq \rho < 1, \quad \mu > 0, \quad \beta > 0.$$

Here ρ is the branching ratio. The stationary mean intensity is

$$\bar{\lambda}(\rho) = \frac{\mu}{1 - \rho}.$$

We use a reduced market-making risk proxy

$$V_{\text{risk}}(\rho) = C(1 - \rho)^{-a}, \quad C > 0,$$

where $a = 2$ matches the normalized-throughput variance convention used for the square-law statement in the main paper. If stationary throughput is modeled separately in the risk term, a can be larger; the proofs keep that dependence explicit.

Assumption A.1 (Local quote sensitivity). *For inventory q in a compact grid and quote controls in a compact interval, the robust half-spread correction is locally linear in the worst-case increase of V_{risk} :*

$$\psi_\varepsilon(\rho) = K \left[\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon(\rho)} V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho) \right] + o \left(\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon(\rho)} |V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho)| \right),$$

with $K > 0$ independent of ρ near one.

This assumption is the scalar version of differentiating the first-order condition for an interior quote optimizer. The optimizer-sensitivity theorem below gives one sufficient route: a uniformly negative quote-space second derivative together with a smooth active worst-case selector for the robust Hamiltonian.

4 Resolvent and Variance Scaling

Lemma A.2 (Scalar resolvent). *For scalar Hawkes branching ratio $\rho < 1$,*

$$(1 - \rho)^{-1}$$

is the scalar Hawkes resolvent. Its square scales as $(1 - \rho)^{-2}$.

Proof. The stationary mean equation is $\bar{\lambda} = \mu + \rho \bar{\lambda}$. Therefore $\bar{\lambda} = (1 - \rho)^{-1} \mu$. The linear response of total descendants of one immigrant is the geometric series $\sum_{n \geq 0} \rho^n = (1 - \rho)^{-1}$, proving the resolvent identity. Squaring the leading response gives $(1 - \rho)^{-2}$. $\square$

Lemma A.3 (Derivative of the risk proxy). *Let $V_{\text{risk}}(\rho) = C(1 - \rho)^{-a}$ for $a > 0$. Then*

$$\frac{d}{d\rho} V_{\text{risk}}(\rho) = aC(1 - \rho)^{-(a+1)}.$$

Proof. Direct differentiation gives $d(1 - \rho)^{-a}/d\rho = a(1 - \rho)^{-(a+1)}$. $\square$

5 Ambiguity Normalizations

We distinguish two ambiguity sets. This is a small modeling choice with a large near-critical consequence, because the normalization changes the singular exponent.

Relative critical-slack ambiguity.

$$\mathcal{U}_\varepsilon^{\text{rel}}(\rho) = \{\tilde{\rho} : 0 \leq \tilde{\rho} \leq \rho + \varepsilon(1 - \rho), \tilde{\rho} < 1\}.$$

Absolute branching-ratio ambiguity.

$$\mathcal{U}_\varepsilon^{\text{abs}}(\rho) = \{\tilde{\rho} : 0 \leq \tilde{\rho} \leq \rho + \varepsilon, \tilde{\rho} < 1\}.$$

Theorem A.4 (Relative-slack amplification). *Fix $a > 0$ and $0 < \varepsilon < 1$. Under relative critical-slack ambiguity,*

$$\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon^{\text{rel}}(\rho)} V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho) = C((1 - \varepsilon)^{-a} - 1)(1 - \rho)^{-a}.$$

If the local quote-sensitivity assumption holds uniformly near the reference regime, then, for small ε ,

$$\psi_\varepsilon(\rho) = KaC\varepsilon(1 - \rho)^{-a} + O(\varepsilon^2(1 - \rho)^{-a}) + o(\varepsilon(1 - \rho)^{-a}).$$

Within this reduced proxy, in the normalized-throughput convention $a = 2$, the premium has the square law $\psi_\varepsilon(\rho) = \Theta(\varepsilon(1 - \rho)^{-2})$.

Proof. The risk proxy is increasing in ρ , so Nature chooses $\tilde{\rho}^* = \rho + \varepsilon(1 - \rho)$. Then

$$1 - \tilde{\rho}^* = 1 - \rho - \varepsilon(1 - \rho) = (1 - \varepsilon)(1 - \rho).$$

Substitution gives

$$V_{\text{risk}}(\tilde{\rho}^*) - V_{\text{risk}}(\rho) = C(1 - \rho)^{-a}((1 - \varepsilon)^{-a} - 1).$$

The Taylor expansion $(1 - \varepsilon)^{-a} = 1 + a\varepsilon + O(\varepsilon^2)$ gives the small- ε expression. The quote-premium statement follows from local quote sensitivity. Since all constants are positive and independent of ρ near one, the result is $\Theta(\varepsilon(1 - \rho)^{-a})$. $\square$

Theorem A.5 (Absolute-ambiguity derivative law). *Fix $a > 0$ and suppose $\varepsilon = o(1 - \rho)$. Under absolute ambiguity,*

$$\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon^{\text{abs}}(\rho)} V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho) = aC\varepsilon(1 - \rho)^{-(a+1)} + O(\varepsilon^2(1 - \rho)^{-(a+2)}).$$

If the local quote-sensitivity assumption holds uniformly near the reference regime, then, within this reduced proxy, the normalized-throughput convention $a = 2$ gives the premium order $\Theta(\varepsilon(1 - \rho)^{-3})$.

Proof. Again Nature chooses $\tilde{\rho}^* = \rho + \varepsilon$ when this remains below one. Taylor expanding around ρ ,

$$V_{\text{risk}}(\rho + \varepsilon) - V_{\text{risk}}(\rho) = \varepsilon V'_{\text{risk}}(\rho) + \frac{\varepsilon^2}{2} V''_{\text{risk}}(\xi)$$

for some $\xi \in (\rho, \rho + \varepsilon)$. By the derivative lemma, $V'_{\text{risk}}(\rho) = aC(1 - \rho)^{-(a+1)}$. Also $V''_{\text{risk}}(\xi) = a(a+1)C(1 - \xi)^{-(a+2)}$. The condition $\varepsilon = o(1 - \rho)$ implies $1 - \xi$ is comparable to $1 - \rho$, giving the stated remainder. Local quote sensitivity transfers the scaling to ψ_ε . $\square$

6 Multitype Perron Extension

Proposition A.6 (First-order Perron perturbation). *Let Γ be nonnegative and irreducible with simple Perron root $\rho < 1$, normalized left/right Perron vectors $\ell^\top r = 1$, spectral gap at least $g_0 > 0$, uniformly bounded Perron eigenprojection, and uniformly bounded reduced resolvent on the non-Perron spectral subspace. Then, for sufficiently small perturbations E ,*

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2/g_0).$$

Among unconstrained Frobenius-unit directions, the first-order maximizer is $E^ = \ell r^\top / \|\ell r^\top\|_F$. Feasibility of a scaled perturbation inside a particular ambiguity set must be checked separately.*

Proof. The first statement is the standard analytic perturbation expansion for a simple eigenvalue. The derivative of the Perron root in direction E is $\ell^\top E r = \langle E, \ell r^\top \rangle_F$. Under a Frobenius norm constraint, Cauchy–Schwarz gives

$$\langle E, \ell r^\top \rangle_F \leq \|E\|_F \|\ell r^\top\|_F,$$

with equality when E is aligned with $\ell r^\top$. $\square$

Remark A.7. *This proposition explains the Perron-aligned adversary used in the experiments when the leading eigenvalue is simple and well separated. It also explains why near-degenerate spectra are a real scope condition rather than a nuisance parameter: the lower bound can weaken when Perron regularity fails.*

Theorem A.8 (Finite-dimensional Perron-visible ambiguity premium). *Let $\Gamma \in \mathbb{R}_+^{d \times d}$ be irreducible, primitive, and subcritical with simple Perron root $\rho < 1$. Assume a uniform spectral gap*

$$\rho - \max_{j>1} |\lambda_j(\Gamma)| \geq g_0 > 0$$

and uniformly bounded Perron eigenprojection and reduced-resolvent conditioning. Let $r, \ell > 0$ be normalized by $\ell^\top r = 1$. For fixed $\mu, w \in \mathbb{R}_+^d$, define

$$V_d(\Gamma) = C \left(w^\top (I - \Gamma)^{-1} \mu \right)^2, \quad C > 0,$$

and assume Perron visibility

$$A(\Gamma) := (w^\top r)(\ell^\top \mu) \geq a_0 > 0.$$

Then

$$V_d(\Gamma) = C A(\Gamma)^2 (1 - \rho)^{-2} + O((1 - \rho)^{-1}).$$

If a feasible Perron-aligned relative-slack perturbation satisfies

$$\rho(\Gamma + E_{\text{rel}}) = \rho + \varepsilon_{\text{rel}}(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho)),$$

and the perturbation path remains in the same compact Perron-regular class with Perron visibility bounded away from zero and no first-order cancellation, then

$$V_d(\Gamma + E_{\text{rel}}) - V_d(\Gamma) = \Theta(\varepsilon_{\text{rel}}(1 - \rho)^{-2}).$$

If a feasible Perron-aligned absolute perturbation satisfies

$$\rho(\Gamma + E_{\text{abs}}) = \rho + \varepsilon + O(\varepsilon^2), \quad \varepsilon = o(1 - \rho),$$

and the same Perron-regularity and visibility conditions remain in force, then

$$V_d(\Gamma + E_{\text{abs}}) - V_d(\Gamma) = \Theta(\varepsilon(1 - \rho)^{-3}).$$

Under the local quote-sensitivity assumption, the robust half-spread premium has the same respective orders.

Proof. By the Riesz projection formula for the isolated Perron eigenvalue,

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho} + R, \quad \|R\| \leq C(g_0),$$

where the constant is uniform on the Perron-regular class. Therefore

$$w^\top (I - \Gamma)^{-1} \mu = \frac{(w^\top r)(\ell^\top \mu)}{1 - \rho} + O(1).$$

Squaring gives the displayed expansion for V_d because Perron visibility keeps the leading coefficient bounded away from zero.

For perturbations, the first-order Perron expansion gives

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2 / g_0).$$

The Perron eigenvectors and the visibility coefficient vary by lower-order terms under the stated local perturbations. Expanding $(1 - \rho - \Delta\rho)^{-2} - (1 - \rho)^{-2}$ gives the relative-slack order when $\Delta\rho = \varepsilon_{\text{rel}}(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho))$, and the absolute-ambiguity derivative law when $\Delta\rho = \varepsilon + O(\varepsilon^2)$ with $\varepsilon = o(1 - \rho)$. The quote-premium statement follows from local quote sensitivity. $\square$

7 Finite-State Robust Bellman Bridge

This section records the finite-state control layer used by the implemented experiment package. It gives the robust dynamic-programming equation for the finite inventory, finite-regime approximation used numerically. By itself, this finite-state theorem is not the continuous-intensity multitype Hawkes HJBI bridge; that compact bridge is stated in the next section.

Let $\mathcal{X}$ be a finite state space whose elements encode inventory, cash-normalized wealth state, and a finite Hawkes-regime or intensity bin. Let $\mathcal{U}(x)$ be a nonempty finite set of admissible quote/no-quote actions at state x , and let $\mathcal{A} = \{\Gamma_1, \dots, \Gamma_M\}$ be a finite rectangular ambiguity set of stable branching matrices. For each $(u, \Gamma) \in \mathcal{U}(x) \times \mathcal{A}$, let $L^{u, \Gamma}(x, y)$ be a conservative generator on $\mathcal{X}$:

$$L^{u, \Gamma}(x, y) \geq 0 \quad (y \neq x), \quad L^{u, \Gamma}(x, x) = - \sum_{y \neq x} L^{u, \Gamma}(x, y).$$

Assume bounded running rewards $r(x, u, \Gamma)$ and bounded terminal payoff $g(x)$.

Theorem A.9 (Finite-state robust Hawkes-regime Bellman verification). *For horizon $T < \infty$, the rectangular robust market-making game*

$$\sup_u \inf_{\Gamma} \mathbb{E} \left[\int_t^T r(X_s, u_s, \Gamma_s) ds + g(X_T) \mid X_t = x \right],$$

where u_s and Γ_s are predictable Markov selectors from $\mathcal{U}(X_s)$ and $\mathcal{A}$, has a value $V(t, x)$. The value is the unique bounded solution of the finite HJBI system

$$-\partial_t V(t, x) = \max_{u \in \mathcal{U}(x)} \min_{\Gamma \in \mathcal{A}} \left[r(x, u, \Gamma) + \sum_{y \in \mathcal{X}} L^{u, \Gamma}(x, y) V(t, y) \right], \quad V(T, x) = g(x).$$

Moreover, deterministic Markov selectors attaining the displayed finite maximum and minimum are optimal. The corresponding discrete-time statement holds for a stochastic one-step kernel; for the explicit Euler kernel this requires Δt small enough that $I + \Delta t L^{u, \Gamma}$ has nonnegative entries and row sums one.

Proof. Because $\mathcal{X}$, $\mathcal{U}(x)$, and $\mathcal{A}$ are finite and rewards are bounded, the Hamiltonian

$$H_x(v) = \max_{u \in \mathcal{U}(x)} \min_{\Gamma \in \mathcal{A}} \left[r(x, u, \Gamma) + \sum_y L^{u, \Gamma}(x, y) v(y) \right]$$

is globally Lipschitz in $v \in \mathbb{R}^{|\mathcal{X}|}$. Indeed, for any v, w , the difference between two generator pairings is bounded by $2\Lambda \|v - w\|_\infty$, where $\Lambda = \max_{x, u, \Gamma} \sum_{y \neq x} L^{u, \Gamma}(x, y) < \infty$; taking finite maxima and minima preserves this Lipschitz constant. Hence the terminal value problem $-\dot{V} = H(V)$, $V(T) = g$, has a unique bounded classical solution by the Picard–Lindelof theorem for finite-dimensional ODEs.

For a small step Δt , conditioning on the first jump of the controlled Markov chain gives

$$V(t, x) = \max_u \min_\Gamma \left[r(x, u, \Gamma) \Delta t + \sum_y P_{\Delta t}^{u, \Gamma}(x, y) V(t + \Delta t, y) \right] + o(\Delta t),$$

where $P_{\Delta t}^{u, \Gamma} = I + \Delta t L^{u, \Gamma} + o(\Delta t)$. Subtract $V(t + \Delta t, x)$, divide by Δt , and let $\Delta t \downarrow 0$ to recover the HJBI system. Conversely, applying Dynkin’s formula to the unique solution under arbitrary Markov selectors and using the pointwise max-min inequality shows that the solution dominates every admissible controller against worst-case Nature and is achieved by any selectors attaining the finite maximum and minimum. Such selectors exist because the action and ambiguity sets are finite. The discrete-time statement follows by the same one-step conditioning argument without taking the limit. $\square$

Remark A.10. *This theorem supports the finite-scenario Bellman layer used in the code. If the scenario set contains Perron-aligned relative-slack perturbations, the finite game can express the robust quote/no-quote response to those regimes. The theorem does not prove the continuum multitype Hawkes HJBI or differentiability of the untruncated optimizer with respect to Γ . The next section gives the compact continuous-intensity bridge.*

8 Compact Continuous-Intensity HJBI Bridge

This section packages the compact-domain verification step, conditional on the stated compact-PDMP dynamic-programming and comparison hypotheses. The argument uses standard PDMP, stochastic-control, and viscosity-solution machinery [3, 2, 1]. The next two sections remove the projection through a Lyapunov/tightness argument and a weighted comparison theorem, and the consolidated theorem below records the resulting untruncated statement.

Let the state be $X_t = (q_t, \lambda_t)$, where $q_t \in \{-Q, \dots, Q\}$ is inventory and $\lambda_t \in K \subset \mathbb{R}_+^d$ is the vector of Hawkes intensities. The compact set K is interpreted as the numerical truncation domain. Between jumps,

$$d\lambda_t = \beta(\mu - \lambda_t) dt.$$

At a mark- i jump, inventory changes by a bounded increment $\zeta_i(q, u)$ and intensity is updated by a truncated Hawkes jump map

$$\lambda_t = \Pi_K(\lambda_{t-} + \beta \Gamma e_i),$$

where Π_K is a fixed Lipschitz projection into K . The market maker chooses $u \in U(q, \lambda)$ and Nature chooses $\Gamma \in \mathcal{A}$, where $U(q, \lambda)$ and $\mathcal{A}$ are compact. Let $\Lambda_i(q, \lambda, u, \Gamma)$ be the controlled mark

intensities after quote placement and censoring by inventory limits. Define the generator on smooth test functions φ by

$$\begin{aligned}\mathcal{L}^{u,\Gamma}\varphi(q,\lambda) &= \beta(\mu - \lambda) \cdot \nabla_\lambda \varphi(q,\lambda) \\ &\quad + \sum_{i=1}^d \Lambda_i(q,\lambda,u,\Gamma) [\varphi(q + \zeta_i(q,u), \Pi_K(\lambda + \beta \Gamma e_i)) - \varphi(q,\lambda)].\end{aligned}$$

Assumption A.11 (Compact Hawkes control regularity). *The sets K , $\mathcal{A}$, and $U(q,\lambda)$ are compact, with $U(q,\lambda)$ having a closed graph. The ambiguity set satisfies $\sup_{\Gamma \in \mathcal{A}} \rho(\Gamma) \leq \rho_{\max} < 1$. The functions $r(q,\lambda,u,\Gamma)$, $g(q,\lambda)$, and $\Lambda_i(q,\lambda,u,\Gamma)$ are bounded and continuous, and are uniformly Lipschitz in (q,λ) on each inventory slice. The jump map above remains inside K , and the total jump rate is uniformly bounded by $\bar{\Lambda}$. Admissible controllers are nonanticipative predictable controls; Nature's choice of Γ is rectangular and predictable over time. The compact PDMP game satisfies the dynamic-programming principle for this admissible class, and bounded viscosity sub/supersolutions of the compact HJBI satisfy the standard comparison principle.*

Theorem A.12 (Conditional compact continuous-intensity Hawkes HJBI verification). *Under compact Hawkes control regularity, the robust finite-horizon rectangular game*

$$V(t,q,\lambda) = \sup_u \inf_{\Gamma} \mathbb{E} \left[\int_t^T r(q_s, \lambda_s, u_s, \Gamma_s) ds + g(q_T, \lambda_T) \mid q_t = q, \lambda_t = \lambda \right]$$

has a bounded uniformly continuous value function. It is the unique bounded viscosity solution, for the compact admissible class specified above, on $[0, T] \times \{-Q, \dots, Q\} \times K$ of

$$-\partial_t V = \sup_{u \in U(q,\lambda)} \inf_{\Gamma \in \mathcal{A}} \left\{ r(q,\lambda,u,\Gamma) + \mathcal{L}^{u,\Gamma} V(q,\lambda) \right\}, \quad V(T,q,\lambda) = g(q,\lambda).$$

If, in addition, the value is C^1 in (t,λ) on each inventory slice and the displayed supremum and infimum admit measurable selectors $u^(t,q,\lambda)$ and $\Gamma^*(t,q,\lambda)$ satisfying Isaacs' pointwise saddle condition, then these selectors verify the game value. In the absence of exact selectors, measurable η -optimizers are η -optimal.*

Proof. The assumptions define a controlled piecewise-deterministic Markov process on the compact state space $\{-Q, \dots, Q\} \times K$. Bounded rates rule out explosion, and compactness with continuity gives measurable selectors for the relaxed one-step Hamiltonians. The assumed compact PDMP dynamic-programming principle gives

$$V(t,x) = \sup_u \inf_{\Gamma} \mathbb{E} \left[\int_t^{t+h} r(X_s, u_s, \Gamma_s) ds + V(t+h, X_{t+h}) \mid X_t = x \right]$$

for $x = (q,\lambda)$.

Let ϕ be a smooth test function touching V from above at (t,x) . Apply the dynamic programming principle with nearly optimal controls over $[t, t+h]$, use the first-jump expansion for the PDMP, subtract $\phi(t,x)$, divide by h , and let $h \downarrow 0$. This gives the viscosity subsolution inequality. The same argument with a test function touching from below, freezing any admissible control and allowing Nature to use a nearly minimizing response, gives the supersolution inequality.

For comparison, the Hamiltonian is proper and uniformly Lipschitz in the value argument because total jump rates are bounded by $\bar{\Lambda}$, rewards are bounded, and coefficients are uniformly Lipschitz on the compact domain. The assumed compact comparison principle therefore implies that

any bounded upper-semicontinuous subsolution is dominated by any bounded lower-semicontinuous supersolution with ordered terminal data. Perron's method and the dynamic programming principle give existence, hence uniqueness of the bounded viscosity solution.

If V is classically differentiable in (t, λ) , applying Dynkin's formula to $V(s, X_s)$ under arbitrary selectors and using the pointwise max-min inequalities proves that V bounds the payoff from above and below. Selectors attaining the saddle Hamiltonian make those inequalities equalities, so they verify the value. Compactness gives measurable η -optimizers when exact selectors are unavailable, yielding η -optimal controls. $\square$

Remark A.13. *This theorem records the compact truncated bridge conditional on the standard compact-PDMP dynamic-programming and comparison hypotheses stated above. The next theorem removes the compact intensity projection for stability and truncation tails. The weighted comparison section then gives the untruncated viscosity identification for pairs with a fixed weighted boundary condition at infinity, and the optimizer-sensitivity section isolates no-quote/boundary active-set differentiability as a separate nonsmooth control issue.*

9 Untruncated Lyapunov Bridge

This section removes the artificial compact intensity projection at the level of nonexplosion, moment control, and truncation error. It is not a universal comparison theorem for every unbounded HJBI; it supplies the missing Lyapunov/tightness component under an explicit common subcriticality condition.

Let $\lambda_t \in \mathbb{R}_+^d$ evolve without projection:

$$d\lambda_t = \beta(\mu - \lambda_t) dt, \quad \lambda_t \mapsto \lambda_t + \beta \Gamma e_i \quad \text{at a mark-}i \text{ jump.}$$

Controls may thin or censor mark intensities, but not amplify them beyond the Hawkes state: assume

$$0 \leq \Lambda_i(q, \lambda, u, \Gamma) \leq \lambda_i$$

for all admissible (q, λ, u, Γ) . This covers passive quote refusal and inventory censoring, which can reduce fill intensities but should not create order-flow arrivals ex nihilo.

Assumption A.14 (Common Lyapunov subcriticality and tail integrability). *There exist $v \in \mathbb{R}_+^d$ with strictly positive entries and $\kappa \in (0, 1]$ such that*

$$\Gamma^\top v \leq (1 - \kappa)v$$

componentwise for every $\Gamma \in \mathcal{A}$. Rewards and terminal payoffs have at most linear Lyapunov growth:

$$|r(q, \lambda, u, \Gamma)| + |g(q, \lambda)| \leq C(1 + v^\top \lambda)$$

uniformly over the finite inventory grid and admissible controls. Fix some $p > 1$. Since $\mathcal{A}$ is compact, the Hawkes jumps have uniformly bounded W_v -increments, so the p th Lyapunov moment calculation below is finite uniformly over $\mathcal{A}$.

Theorem A.15 (Conditional untruncated Hawkes Lyapunov bridge). *Under common Lyapunov subcriticality, the unprojected controlled Hawkes PDMP is nonexplosive for every predictable admissible controller and Nature selector. For*

$$W_v(\lambda) = 1 + v^\top \lambda,$$

its generator satisfies the uniform drift bound

$$\mathcal{L}^{u,\Gamma} W_v(q, \lambda) \leq \beta v^\top \mu - \beta \kappa v^\top \lambda \leq C_0 - C_1 W_v(\lambda)$$

for constants $C_0, C_1 > 0$ depending only on (β, μ, v, κ) . Consequently, for every finite horizon T and the $p > 1$ in the assumption,

$$\sup_{u,\Gamma} \mathbb{E}_{q,\lambda} \left[\sup_{0 \leq s \leq T} W_v(\lambda_s)^p \right] \leq C_{T,p} W_v(\lambda)^p,$$

and the finite-horizon value is finite in the weighted class $|V(t, q, \lambda)| \leq C_T W_v(\lambda)$. Moreover, if $\tau_R = \inf\{t : W_v(\lambda_t) \geq R\}$, then

$$\sup_{u,\Gamma} \mathbb{P}_{q,\lambda}(\tau_R \leq T) \leq \frac{C_{T,p} W_v(\lambda)^p}{R^p}.$$

Thus compact truncations on $\{W_v \leq R\}$ approximate the untruncated game in probability. For linearly growing payoffs, the exit-tail contribution is uniformly controlled by

$$\sup_{u,\Gamma} \mathbb{E}_{q,\lambda} \left[\sup_{s \leq T} W_v(\lambda_s) \mathbf{1}_{\{\tau_R \leq T\}} \right] \leq \frac{C_{T,p} W_v(\lambda)^p}{R^{p-1}},$$

so the compact-truncation payoff error vanishes locally uniformly as $R \uparrow \infty$.

Proof. For the affine test function W_v , the deterministic drift contributes

$$\beta(\mu - \lambda) \cdot \nabla_\lambda W_v = \beta v^\top \mu - \beta v^\top \lambda.$$

At a mark- i jump, W_v increases by $\beta v^\top \Gamma e_i$. Therefore

$$\mathcal{L}^{u,\Gamma} W_v = \beta v^\top \mu - \beta v^\top \lambda + \sum_{i=1}^d \Lambda_i(q, \lambda, u, \Gamma) \beta v^\top \Gamma e_i.$$

Using $\Lambda_i \leq \lambda_i$ and the common Lyapunov inequality gives

$$\sum_i \Lambda_i v^\top \Gamma e_i \leq \sum_i \lambda_i (\Gamma^\top v)_i \leq (1 - \kappa) \sum_i \lambda_i v_i.$$

Substitution yields

$$\mathcal{L}^{u,\Gamma} W_v \leq \beta v^\top \mu - \beta \kappa v^\top \lambda.$$

Since $W_v = 1 + v^\top \lambda$, this is the stated Foster–Lyapunov drift after absorbing constants.

Dynkin’s formula applied to $W_v(\lambda_{t \wedge \tau_R})$ gives a uniform first-moment bound up to τ_R . Gronwall’s inequality gives $\mathbb{E} W_v(\lambda_{t \wedge \tau_R}) \leq C_T W_v(\lambda_0)$ with constants independent of R , u , and Γ . Letting $R \uparrow \infty$ proves nonexplosion and the first-moment estimate.

For the p th moment, apply the generator to W_v^p . Each mark- i jump adds the bounded amount $c_i(\Gamma) = \beta v^\top \Gamma e_i$ to W_v . Taylor’s formula gives, for $W_v \geq 1$,

$$(W_v + c_i)^p - W_v^p \leq p c_i W_v^{p-1} + C_p W_v^{p-2}.$$

The deterministic drift contributes $p W_v^{p-1} \beta(v^\top \mu - v^\top \lambda)$, while

$$\sum_i \lambda_i c_i(\Gamma) \leq \beta(1 - \kappa) v^\top \lambda.$$

Thus, uniformly over controls and Nature,

$$\mathcal{L}^{u,\Gamma} W_v^p \leq C_p W_v^{p-1} - p\beta\kappa W_v^p \leq C'_p - c_p W_v^p$$

after increasing constants. Dynkin's formula and Gronwall yield $\mathbb{E}W_v(\lambda_{t\wedge\tau_R})^p \leq C_{T,p}W_v(\lambda_0)^p$ uniformly in R , u , and Γ .

For the maximal estimate, write the stopped semimartingale decomposition

$$W_v(\lambda_{t\wedge\tau_R}) = W_v(\lambda_0) + A_{t\wedge\tau_R} + M_{t\wedge\tau_R}.$$

The drift variation satisfies $|A_{t\wedge\tau_R}| \leq C \int_0^{t\wedge\tau_R} W_v(\lambda_s) ds$. The predictable quadratic variation of M is also bounded by $C \int_0^{t\wedge\tau_R} W_v(\lambda_s) ds$, because each mark- i jump changes W_v by the fixed amount $\beta v^\top \Gamma e_i$ and the total marked jump rate is at most $\sum_i \lambda_i$. Burkholder–Davis–Gundy with exponent p and Gronwall then give

$$\mathbb{E} \sup_{s \leq T} W_v(\lambda_{s\wedge\tau_R})^p \leq C_{T,p} W_v(\lambda_0)^p,$$

uniformly in R . Letting $R \uparrow \infty$ proves the displayed maximal moment bound. Markov's inequality then gives the exit bound for τ_R .

The value-growth statement follows by integrating the assumed linear growth of running rewards and terminal payoff against the moment bound. On $\{W_v \leq R\}$, the compact HJBI theorem applies to the truncated game. The p th maximal moment and Holder's inequality give

$$\mathbb{E} \left[\sup_{s \leq T} W_v(\lambda_s) \mathbf{1}_{\{\tau_R \leq T\}} \right] \leq \left(\mathbb{E} \sup_{s \leq T} W_v(\lambda_s)^p \right)^{1/p} \mathbb{P}(\tau_R \leq T)^{1-1/p},$$

which is bounded by $C_{T,p} W_v(\lambda_0)^p / R^{p-1}$. This controls the payoff contribution from paths that leave the truncation set before T , proving convergence in probability and tail-controlled approximation for linearly growing payoffs. $\square$

Remark A.16. *If $\mathcal{A}$ is finite and each matrix is strictly subcritical, the common vector condition is easy to check by linear programming. For a general compact ambiguity set, it is a substantive robust-stability assumption, stronger than pointwise $\rho(\Gamma) < 1$. This is the tradeoff required by a uniform untruncated theorem.*

Remark A.17. *Combining the compact HJBI theorem with the Lyapunov bridge gives an untruncated value approximation theorem in the weighted class: solve on $\{W_v \leq R\}$, stop at the boundary, and let $R \uparrow \infty$. The next section adds the corresponding viscosity comparison theorem in the W_v -weighted class with an explicit boundary condition at infinity.*

10 Weighted Comparison and Global Untruncated HJBI

This section states a sufficient comparison condition for the unbounded-domain HJBI bridge, following the standard viscosity comparison strategy with a weighted barrier and a doubling-of-variables argument [1, 2]. It is deliberately explicit: without a weighted boundary condition at infinity, unbounded intensity states can admit spurious solutions.

Assumption A.18 (Weighted comparison regularity). *In addition to common Lyapunov subcriticality, assume:*

1. The ambiguity set $\mathcal{A}$ is compact, and the control set is either a fixed compact set or a compact correspondence $U(q, \lambda)$ continuous in Hausdorff distance on compact λ -sets; measurable η -selectors exist for the local Hamiltonian.

2. The controlled rates $\Lambda_i(q, \lambda, u, \Gamma)$ are continuous, bounded above by λ_i , and locally Lipschitz in λ , uniformly on compact subsets and uniformly over (u, Γ) .

3. The jump maps

$$J_i(q, \lambda, u, \Gamma) = (q + \zeta_i(q, u), \lambda + \beta \Gamma e_i)$$

are Lipschitz in λ and satisfy

$$W_v(J_i(q, \lambda, u, \Gamma)) \leq C_J W_v(\lambda).$$

4. Rewards and terminal payoffs are continuous and have at most linear W_v -growth.

5. Subsolutions U and supersolutions V are compared in the weighted class

$$|U(t, q, \lambda)| + |V(t, q, \lambda)| \leq C W_v(\lambda)$$

and satisfy the boundary condition at infinity

$$\limsup_{R \rightarrow \infty} \sup_{\substack{t \in [0, T], q \\ W_v(\lambda) \geq R}} \frac{U(t, q, \lambda) - V(t, q, \lambda)}{W_v(\lambda)} \leq 0.$$

Theorem A.19 (Sufficient weighted-comparison bridge for the untruncated Hawkes HJBI). *Under weighted comparison regularity, let U be an upper-semicontinuous viscosity subsolution and V a lower-semicontinuous viscosity supersolution of the untruncated robust Hawkes HJBI*

$$-\partial_t F = \sup_{u \in U(q, \lambda)} \inf_{\Gamma \in \mathcal{A}} \left\{ r(q, \lambda, u, \Gamma) + \mathcal{L}^{u, \Gamma} F(q, \lambda) \right\}$$

on $[0, T] \times \{-Q, \dots, Q\} \times \mathbb{R}_+^d$. If $U(T, q, \lambda) \leq V(T, q, \lambda)$ for all (q, λ) and the weighted boundary condition above holds, then

$$U(t, q, \lambda) \leq V(t, q, \lambda)$$

for every (t, q, λ) . Consequently, any two candidates in the same weighted-boundary subclass are ordered by comparison. Any compact-truncation limit satisfying this boundary condition is therefore uniquely identified, and the truncated solutions converge locally uniformly to it as $R \uparrow \infty$.

Proof. Fix $\delta > 0$ and define the strict supersolution

$$V^\delta(t, q, \lambda) = V(t, q, \lambda) + \delta e^{a(T-t)} W_v(\lambda),$$

with a chosen large enough that the Lyapunov drift dominates the linear growth constants in the Hamiltonian. The Foster–Lyapunov inequality gives

$$-\partial_t \{\delta e^{a(T-t)} W_v\} - \sup_u \inf_\Gamma \mathcal{L}^{u, \Gamma} \{\delta e^{a(T-t)} W_v\} \geq c \delta W_v(\lambda) - C \delta$$

for some $c > 0$ after increasing a . Thus V^δ is a strict supersolution outside a fixed compact set, and the weighted boundary condition implies that any positive maximum of $U - V^\delta$ occurs on $[0, T] \times \{-Q, \dots, Q\} \times K_R$ for some compact Lyapunov sublevel set $K_R = \{W_v \leq R\}$.

On this compact set, apply the standard doubling-of-variables argument to

$$U(t, q, \lambda) - V^\delta(s, q, \lambda') - \frac{|t - s|^2}{2\alpha} - \frac{|\lambda - \lambda'|^2}{2\varepsilon}.$$

The finite inventory coordinate is discrete, so at a positive maximum it can be matched by taking the same q ; otherwise the maximum over finitely many inventory states is handled componentwise. The viscosity inequalities at the maximizers, together with compactness of $U(q, \lambda)$ and $\mathcal{A}$, give nearly maximizing/minimizing controls and ambiguity selectors. Continuity and local Lipschitz regularity of rates, rewards, and jump maps control the drift and jump terms. The nonlocal terms are handled by the viscosity nonlocal doubling inequality:

$$U(t, J_i(q, \lambda, u, \Gamma)) - V^\delta(s, J_i(q, \lambda', u, \Gamma))$$

is bounded by the same doubled maximum plus an $O(|\lambda - \lambda'|^2/\varepsilon) + o(1)$ error, because each J_i is Lipschitz on K_R . Letting $\alpha, \varepsilon \downarrow 0$ yields a contradiction to strict positivity of the maximum unless $\sup(U - V^\delta) \leq 0$.

Therefore $U \leq V^\delta$ for every $\delta > 0$. Letting $\delta \downarrow 0$ proves comparison. Existence follows by solving the compactly truncated HJBI on K_R , extending with stopped dynamics, and using the Lyapunov exit estimate from the previous theorem to pass to the limit. Comparison identifies every subsequential limit satisfying the same weighted boundary condition at infinity, so the convergence is locally uniform and the limit is the unique viscosity solution in that fixed-boundary subclass. $\square$

Remark A.20. *This theorem records the untruncated HJBI comparison bridge for the model class in this appendix: common Lyapunov subcriticality with $p > 1$ tail control, linear-growth rewards, compact controls/ambiguity, and a fixed weighted boundary behavior at infinity. It still does not prove differentiability of boundary or no-quote active sets, nor does it cover state-dependent, power-law, or empirically learned Hawkes kernels without rechecking the same weighted comparison assumptions.*

11 Interior Quote Optimizer Sensitivity

The HJBI theorems above identify the value for the stated control class. This section records the additional local condition needed to connect the robust Hamiltonian to differentiable quote rules as the branching matrix or ambiguity parameter changes. It covers the smooth interior region of the implemented quote map; no-quote boundaries and active-set switches are intentionally handled as nonsmooth cases.

Fix a state/adjoint tuple z collecting (t, q, λ) and the value increments entering the local HJBI Hamiltonian. Let $\theta \in \Theta \subset \mathbb{R}^m$ be a finite-dimensional parameterization of either Γ itself or an ambiguity set centered at Γ . For a quote vector $\delta \in D \subset \mathbb{R}^k$, define the reduced robust Hamiltonian

$$\mathfrak{H}(\delta, \theta; z) = \inf_{\Gamma' \in \mathcal{A}(\theta)} \mathcal{H}(\delta, \Gamma'; z),$$

where $\mathcal{H}$ is the local reward-plus-generator expression in the HJBI.

Assumption A.21 (Smooth interior robust optimizer). *For the fixed z , assume there are neighborhoods of (δ_0, θ_0) such that:*

1. D is open around δ_0 , and the admissible quote constraints are inactive in that neighborhood.
2. Nature has a unique active minimizer $\Gamma^\dagger(\delta, \theta; z)$, and this selector is C^1 in (δ, θ) .

3. The reduced Hamiltonian $\mathfrak{H}$ is C^2 in δ and has continuous mixed derivative $D_{\delta\theta}^2 \mathfrak{H}$.

4. The quote objective is uniformly strictly concave:

$$D_{\delta\delta}^2 \mathfrak{H}(\delta, \theta; z) \preceq -mI_k$$

for some $m > 0$ throughout the neighborhood.

5. δ_0 is the unique local maximizer at θ_0 , so $D_\delta \mathfrak{H}(\delta_0, \theta_0; z) = 0$.

Theorem A.22 (Interior optimizer differentiability). *Under the smooth interior robust optimizer assumption, there exists a neighborhood N of θ_0 and a unique interior maximizer $\delta^*(\theta; z)$ in the corresponding quote neighborhood. The map $\theta \mapsto \delta^*(\theta; z)$ is C^1 on N , and*

$$D_\theta \delta^*(\theta_0; z) = - \left[D_{\delta\delta}^2 \mathfrak{H}(\delta_0, \theta_0; z) \right]^{-1} D_{\delta\theta}^2 \mathfrak{H}(\delta_0, \theta_0; z).$$

In the one-dimensional half-spread case, if e is a Perron-adverse structural direction and

$$D_{\delta\theta}^2 \mathfrak{H}(\delta_0, \theta_0; z)e \geq 0,$$

then $D_\theta \delta^*(\theta_0; z)e \geq 0$; the robust interior half-spread widens to first order in that adverse direction.

Proof. By the smooth active-minimizer assumption and the envelope/Danskin theorem, the reduced Hamiltonian has the stated derivatives in a neighborhood of (δ_0, θ_0) . Define

$$G(\delta, \theta) = D_\delta \mathfrak{H}(\delta, \theta; z).$$

The first-order condition at the reference point is $G(\delta_0, \theta_0) = 0$. The derivative of G with respect to δ is $D_{\delta\delta}^2 \mathfrak{H}(\delta_0, \theta_0; z)$, which is nonsingular because it is negative definite. The implicit function theorem therefore gives a unique C^1 function $\delta^*(\theta; z)$ solving $G(\delta^*(\theta; z), \theta) = 0$ locally. Strict concavity makes that critical point the unique local maximizer. Differentiating the identity $G(\delta^*(\theta; z), \theta) = 0$ gives the displayed formula. In dimension one, the denominator $D_{\delta\delta}^2 \mathfrak{H}$ is negative, so the sign of the directional derivative agrees with the sign of $D_{\delta\theta}^2 \mathfrak{H}e$. This proves the widening statement. $\square$

Remark A.23. *The theorem is local by design. If the optimal quote hits a no-quote boundary, if a cap such as $\delta \leq \delta_{\max}$ binds, or if two worst-case Hawkes matrices exchange activity, the optimizer may have only one-sided directional derivatives or a kink. Those cases are consistent with the implemented liquidity-guard policy, but they should be reported as nonsmooth active-set behavior rather than as classical differentiability.*

12 Consolidated Multitype Control Theorem

The preceding sections are intentionally modular because compact verification, untruncated stability, comparison, and quote sensitivity require different assumptions. The theorem below gives a single citation point for the sufficient conditions assembled above.

Theorem A.24 (Consolidated conditional multitype Hawkes control bridge). *Consider the finite-horizon robust market-making game with finite inventory grid, exponential d -type Hawkes intensity state $\lambda \in \mathbb{R}_+^d$, compact quote/no-quote control graph, compact rectangular ambiguity set $\mathcal{A}$, controlled mark rates bounded by the Hawkes intensity state, and continuous rewards and terminal payoff with at most linear W_v -growth. Suppose common Lyapunov subcriticality with the $p > 1$ tail-integrability condition and weighted comparison regularity hold, and compare candidate viscosity solutions only in a W_v -weighted subclass with the same stated boundary condition at infinity. Then:*

1. the controlled Hawkes PDMP is nonexplosive for every admissible controller and Nature selector;
2. the robust finite-horizon value is finite in the W_v -weighted class;
3. the value is the unique viscosity solution, in that weighted class and with the fixed stated boundary condition at infinity, of the untruncated multitype Hawkes HJBI equation;
4. compact Lyapunov-sublevel truncations converge locally uniformly to the untruncated value as the truncation radius tends to infinity;
5. if the untruncated value is classically differentiable, has the same W_v -growth, its generator is integrable under the selected localized dynamics, and the Isaacs saddle selectors exist with admissible localization limits, then those selectors verify the untruncated game value; and
6. on smooth interior quote regions satisfying the smooth interior robust optimizer assumption, with inactive quote constraints, a unique C^1 active worst-case structural selector, and a C^2 reduced Hamiltonian with continuous mixed derivative, uniformly negative quote Hessian, and a unique local maximizer—the robust quote map is C^1 in the branching/ambiguity parameter and its derivative is given by the implicit-function formula above.

Proof. Items 1 and 2 follow from the Lyapunov stability theorem: the common subcritical vector gives the Foster–Lyapunov drift bound, nonexplosion, $p > 1$ finite weighted maximal moments, and finite linearly growing payoffs. Item 3 follows from the weighted comparison theorem, which proves comparison for upper-semicontinuous subsolutions and lower-semicontinuous supersolutions in the W_v -weighted class with the boundary condition at infinity; existence is obtained by the compact truncation construction. The boundary condition at infinity is a condition on the compared candidate solutions, not a consequence of pointwise subcriticality alone. Item 4 is the local-uniform convergence conclusion of the same comparison-plus-truncation argument, using stopped Lyapunov-sublevel games and the Lyapunov exit estimate to control paths that leave compact sublevel sets; it does not identify projected-boundary controls with stopped-boundary controls without this localization error bound. Item 5 is the classical verification argument localized at τ_R , followed by Dynkin’s formula, the max–min inequalities, the weighted moment bounds, and then $R \uparrow \infty$ under the stated generator integrability and selector localization assumptions. Item 6 is the interior optimizer differentiability theorem, with its smooth active-minimizer and strict concavity hypotheses included explicitly in the theorem statement. $\square$

13 Bivariate Perron Bridge

This section records the bivariate bridge theorem used by the manuscript. It is still a reduced-form risk-premium theorem, not a complete HJBI proof, but it upgrades the scalar statement to the smallest genuinely multitype Hawkes setting.

Let

$$\Gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}$$

be nonnegative, irreducible, and subcritical with Perron root $\rho = \rho(\Gamma) < 1$. Let $r, \ell > 0$ be right/left Perron vectors normalized by $\ell^\top r = 1$. Let the second eigenvalue be η , and assume a fixed gap condition along a near-critical sequence,

$$\rho \uparrow 1, \quad |\rho - \eta| \geq g_0 > 0.$$

For two-type order flow with baseline $\mu \in \mathbb{R}_+^2$, the stationary intensity is

$$\bar{\lambda} = (I - \Gamma)^{-1}\mu.$$

For a reduced market-making risk direction $w \in \mathbb{R}_+^2$, define

$$V_{\text{bi}}(\Gamma) = C \left(w^\top (I - \Gamma)^{-1} \mu \right)^2, \quad C > 0.$$

We work on a compact Perron-regular class with uniformly bounded eigenprojection conditioning and Perron visibility bounded below:

$$A := (w^\top r)(\ell^\top \mu) \geq a_0 > 0.$$

Lemma A.25 (Bivariate resolvent decomposition). *Under the bivariate assumptions above,*

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho} + R_\eta, \quad \|R_\eta\| \leq C_\eta,$$

where C_η depends only on the spectral gap, vector normalization, eigenprojection conditioning, and a compact upper bound on $\|\Gamma\|$.

Proof. Because Γ is a real 2×2 matrix with simple Perron root and separated second eigenvalue, the spectral projection onto the Perron eigenspace is $P = r\ell^\top$. The complementary projection is $I - P$. On the Perron subspace, $(I - \Gamma)^{-1}$ has eigenvalue $(1 - \rho)^{-1}$. On the complementary one-dimensional subspace, the resolvent has eigenvalue $(1 - \eta)^{-1}$ times a bounded projection. Since $|\rho - \eta| \geq g_0$ and $\rho < 1$, the complementary term is uniformly bounded on the stated compact class. This gives the decomposition. $\square$

Lemma A.26 (Bivariate risk singularity). *Under the same assumptions,*

$$V_{\text{bi}}(\Gamma) = CA^2(1 - \rho)^{-2} + O((1 - \rho)^{-1}).$$

Consequently $V_{\text{bi}}(\Gamma) = \Theta((1 - \rho)^{-2})$ as $\rho \uparrow 1$.

Proof. By the resolvent decomposition,

$$w^\top (I - \Gamma)^{-1} \mu = \frac{(w^\top r)(\ell^\top \mu)}{1 - \rho} + w^\top R_\eta \mu = \frac{A}{1 - \rho} + O(1).$$

Squaring yields

$$\left(\frac{A}{1 - \rho} + O(1) \right)^2 = A^2(1 - \rho)^{-2} + O((1 - \rho)^{-1}) + O(1).$$

Since $A > 0$, the leading term is nonzero and proves the stated singularity. $\square$

Assumption A.27 (Perron-coefficient regularity). *For the local perturbation paths below, the Perron visibility coefficient*

$$A(\Gamma) := (w^\top r(\Gamma))(\ell(\Gamma)^\top \mu)$$

does not cancel the gap singularity. Concretely, for relative-slack paths we assume $A(\Gamma + E_{\text{rel}}) = A(\Gamma) + o(\varepsilon_{\text{rel}})$, and for absolute paths we assume $A(\Gamma + E_{\text{abs}}) = A(\Gamma) + o(\varepsilon)$. A sufficient special case is a spectral path that changes the Perron root while holding the Perron eigendirections fixed to first order.

Theorem A.28 (Perron-aligned bivariate relative-slack premium). *Let E_{rel} be a perturbation for which*

$$\rho(\Gamma + E_{\text{rel}}) = \rho + \varepsilon_{\text{rel}}(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho)),$$

and suppose $\Gamma + E_{\text{rel}}$ remains in the same compact bivariate Perron-regular class with Perron visibility bounded below and satisfying Perron-coefficient regularity. Then

$$V_{\text{bi}}(\Gamma + E_{\text{rel}}) - V_{\text{bi}}(\Gamma) = 2CA^2\varepsilon_{\text{rel}}(1 - \rho)^{-2} + O(\varepsilon_{\text{rel}}^2(1 - \rho)^{-2}) + o(\varepsilon_{\text{rel}}(1 - \rho)^{-2}) + O(\varepsilon_{\text{rel}}(1 - \rho)^{-1}).$$

In particular, the leading singular order is $\Theta(\varepsilon_{\text{rel}}(1 - \rho)^{-2})$. Under local quote sensitivity, if the finite ambiguity set includes this Perron-aligned scenario and that scenario is selected as worst case, then the half-spread has the same square-law order.

Proof. The bivariate risk singularity gives

$$V_{\text{bi}}(\Gamma) = CA^2(1 - \rho)^{-2} + O((1 - \rho)^{-1}).$$

The perturbed gap satisfies

$$1 - \rho(\Gamma + E_{\text{rel}}) = (1 - \varepsilon_{\text{rel}})(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho)).$$

Uniform Perron visibility and compactness keep the leading coefficient bounded above and below. Therefore the leading change is the same calculation as in the scalar relative-slack theorem with $a = 2$:

$$CA^2(1 - \rho)^{-2}\{(1 - \varepsilon_{\text{rel}})^{-2} - 1\} = 2CA^2\varepsilon_{\text{rel}}(1 - \rho)^{-2} + O(\varepsilon_{\text{rel}}^2(1 - \rho)^{-2}).$$

The lower bound follows because A is bounded away from zero and the Perron-coefficient regularity prevents first-order cancellation; the complementary resolvent term is lower order. Local quote sensitivity transfers the order to the half-spread correction. $\square$

Theorem A.29 (Perron-aligned bivariate absolute-ambiguity derivative law). *Let E_{abs} satisfy*

$$\rho(\Gamma + E_{\text{abs}}) = \rho + \varepsilon + O(\varepsilon^2), \quad \varepsilon = o(1 - \rho),$$

and suppose the perturbed matrices remain in the same bivariate Perron-regular class with Perron visibility bounded below and satisfying Perron-coefficient regularity. Then

$$V_{\text{bi}}(\Gamma + E_{\text{abs}}) - V_{\text{bi}}(\Gamma) = 2CA^2\varepsilon(1 - \rho)^{-3} + O(\varepsilon^2(1 - \rho)^{-4}) + o(\varepsilon(1 - \rho)^{-3}) + O(\varepsilon(1 - \rho)^{-2}).$$

In particular, the leading singular order is $\Theta(\varepsilon(1 - \rho)^{-3})$. Under local quote sensitivity, if the finite ambiguity set includes this Perron-aligned scenario and that scenario is selected as worst case, then the half-spread has the same derivative-law order.

Proof. Using the same leading expansion,

$$V_{\text{bi}}(\Gamma + E_{\text{abs}}) - V_{\text{bi}}(\Gamma) = CA^2 \left((1 - \rho - \varepsilon + O(\varepsilon^2))^{-2} - (1 - \rho)^{-2} \right) + O(\varepsilon(1 - \rho)^{-2}).$$

Since $\varepsilon = o(1 - \rho)$, a Taylor expansion of x^{-2} at $x = 1 - \rho$ gives

$$(1 - \rho - \varepsilon)^{-2} - (1 - \rho)^{-2} = 2\varepsilon(1 - \rho)^{-3} + O(\varepsilon^2(1 - \rho)^{-4}).$$

The $O(\varepsilon(1 - \rho)^{-2})$ contribution from the bounded complementary resolvent and the $o(\varepsilon(1 - \rho)^{-3})$ contribution from the Perron coefficient are lower order. Positivity of the aligned Perron coefficient gives the matching lower bound. Local quote sensitivity again transfers the order. $\square$

Remark A.30 (How to realize the perturbations). *The first-order Perron perturbation proposition gives a concrete construction: take E proportional to the Frobenius-normalized Perron direction $\ell r^\top$. Then*

$$\rho(\Gamma + tE) - \rho(\Gamma) = t\langle E, \ell r^\top \rangle_F + O(t^2),$$

so choosing $t = O(\varepsilon_{\text{rel}}(1 - \rho))$ gives the relative theorem and choosing $t = O(\varepsilon)$ gives the absolute theorem. This matches the Perron-aligned direction used in the matrix-resolvent experiments, up to the feasibility and normalization choices imposed there.

14 Conclusion

The square-law statement should be read with its normalization in view. When the ambiguity radius is measured relative to the remaining critical slack, or when the risk quantity has already been normalized, the reduced premium has the square-law order. When ε is an absolute branching-matrix radius, the robust premium inherits one additional derivative of the critical singularity. The same distinction holds in the bivariate Perron-visible setting. In that sense, this appendix puts the MESA claim into precise, testable scalar and bivariate theorem statements.

References

- [1] M. G. Crandall, H. Ishii, and P.-L. Lions. User’s guide to viscosity solutions of second order partial differential equations. *Bulletin of the American Mathematical Society*, 27(1):1–67, 1992. doi:10.1090/S0273-0979-1992-00266-5.
- [2] W. H. Fleming and H. M. Soner. *Controlled Markov Processes and Viscosity Solutions*. 2nd ed., Springer, 2006. doi:10.1007/0-387-31071-1.
- [3] M. H. A. Davis. *Markov Models and Optimization*. Chapman & Hall, 1993. ISBN 9780412314100.